



Software Release Notes for the Mobile Content Cloud

Release 9.4.0.0

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Release Abstract

This document contains the software release notes for the Affirmed Networks Mobile Content Cloud, Release 9.4.0.0. This version of MCC operates with other released products as follows:

- Acuitas version 9.4.0.0
- dpisig_2019-10-18.tgz

For more information see the *Software Release Notes for the Acuitas Service Management System* and the *DPI Classification Release Notes for R8.4 and above*.

Bundled Software in This Release

The following files are included with the software in this release:

- Debian software package file list.

This file uses the following file naming convention:

an-`<release-number>`-`<build number>`.REL-packages.txt

For example, *an-9.4.0.0-455.REL-packages.txt*.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the Adobe Reader®. To download the latest version of the Adobe Reader software from the Adobe web site, go to <http://get.adobe.com/reader/>

Product documentation is available from the Affirmed Networks SFTP server. Contact your sales representative or the TAC for instructions on accessing the SFTP server.

See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

Technical Assistance Center

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact the TAC:



E-mail: prodsupport@affirmednetworks.com



24 x 7 Telephone support:

Local: 1-978-268-0871

Toll free: 1-888-283-2838

Virtual Platform Minimum Software Requirements

The following minimum virtual platform software versions are qualified by Affirmed Networks for use with this release. If you are running a different version of software, contact the TAC for assistance.

Module	Minimum Version or Release
Enclosure Level:	
Onboard Administrator	4.50
HP VC FlexFabric 10 Gb / 24 Port Module	4.45
HP 6125XLG Blade Switch	2422P01
BL460C Blade:	
System ROM	Gen 8: 2015.06.01 Gen 9: 2.00_12-28-2015
iLO	2.40
Power Management Controller	Gen 8: 3.30 Gen 9: 1.09
Smart Array P220i	Gen 8: Smart array P220i controller 7.02 Gen 9: Smart array P244br controller 3.56

Third-Party Software Requirements

The following third-party software versions are qualified by Affirmed Networks for use with this release. If you are running a different version of third-party software, contact the TAC for assistance.

[Table 1](#) lists the VMware configuration software requirements for each hardware platform.

Table 1 VMware Configuration

Hardware Platform	VMware Configuration						
	VIMs	Hypervisor	Switching Technology ^a	NIC	Firmware Version	Host Driver	ISO Image Name
Dell R740	vCenter 6.7.0	VMware ESXi 6.7, Update 2	SR-IOV only	Fortville 25G (Intel)	6.80 0x80003d17 18.8.9	i40en version 1.7.17	VMware-VMvisor-Installer-6.7.0.update02-13006603.x86_64.iso
Dell R740	vCenter 6.5.0	VMware ESXi 6.5, Update 2	SR-IOV only	Fortville 25G (Intel)	6.80 0x80003d17 18.8.9	i40en 1.7.11-1OEM.650.0.0.4598673	VMware-VMvisor-Installer-6.5.0.update02-10719125.x86_64-DellEMC_Customized-A07.iso
HP DL380 Gen 10/ Gen 9	vCenter 6.5.0	VMware ESXi 6.5, Update 2	SR-IOV only	Mellanox 25G	14.23.244	nmlx5_core 4.16.14.2	an-v2.0-VMware-ESXi-6.5.0-Update2-HPE-Gen9Plus-650.U2.10.3.5.5.iso
HP DL380 Gen 9	vCenter 6.0.0	VMware ESXi 6.0, Update 2	vSwitch, vDS	Niantic 10G (Intel)	-	ixgben 1.6.5-1OEM.600.0.0.2768847	an-VMware-ESXi-6.0.0-Update2-3620759-HPE-600-9.5.0.48-Apr2016.iso
				Fortville 10G (Intel)	-	INT_bootbank_i40en_1.7.11-1OEM.600.0.0.2768847.vib	
HP c7000 Gen 9	vCenter 6.5.0	VMware ESXi 6.5, Update 2	SR-IOV only	Niantic 10G (Intel)	0x8000083d	ixgben 1.6.5-1OEM.600.0.0.2768847	an-v2.0-VMware-ESXi-6.5.0-Update2-HPE-Gen9Plus-650.U2.10.3.5.5.iso
HP c7000 Gen 9	vCenter 6.0.0	VMware ESXi 6.0, Update 2	vSwitch, vDS	Niantic 10G (Intel)	-	ixgben 1.6.5-1OEM.600.0.0.2768847	an-VMware-ESXi-6.0.0-Update2-3620759-HPE-600-9.5.0.48-Apr2016.iso

^a SR-IOV is supported on all platforms.

Table 2 lists the KVM configuration software requirements for each hardware platform.

Table 2 KVM Configuration

Hardware Platform	KVM Configuration						
	VIMs	Operating System	Switching Technology ^a	NIC	Firmware Version	Host Driver	ISO Image Name
HP DL380 Gen 9	virt-manager	RedHat 7.6	Linux Bridge	Niantic 10G (Intel)	0x80000886, 1.1752.0	ixgbe 5.1.0-k-rh7.6	an-v1.0-osp13-rhel-server-7.6-x86_64.iso
HP DL380 Gen 9	virt-manager	RedHat 7.4	Linux Bridge	Fortville 10G (Intel)	6.80 0x80003d17 18.8.9	140e-2.4.6.tar.gz	an-v1.6-osp10-rhel-server-7.4-x86_64.iso
HP DL380 Gen 9	virt-manager	RedHat 7.3	Linux Bridge	Mellanox 10G	-	MLNX_OFED_LINUX-4.3-1.0.1.0-rhel7.3-x86_64.iso	an-v1.5-osp10-rhel-server-7.3-x86_64.iso
HP c7000 Gen 9	virt-manager	RedHat 7.6	Linux Bridge	Niantic 10G (Intel)	0x80000886, 1.1752.0	ixgbe 5.1.0-k-rh7.6	an-v1.0-osp13-rhel-server-7.6-x86_64.iso
Dell R630	virt-manager	RedHat 7.3	Linux Bridge	Mellanox 10G	-	MLNX_OFED_LINUX-4.3-1.0.1.0-rhel7.3-x86_64.iso	an-v1.5-osp10-rhel-server-7.3-x86_64.iso

^a The supported switching technology applies to base and management interfaces. SR-IOV is supported on N/S and E/W interfaces.

Table 3 lists the OpenStack configuration software requirements for each hardware platform.

Table 3 OpenStack Configuration

Hardware Platform	OpenStack Configuration						
	VIMs	Operating System	Switching Technology ^a	NIC	Firmware Version	Host Driver	ISO Image Name
Dell R740	OpenStack Queens	RedHat 7.7	■ SR-IOV ■ OVS	Fortville 25G (Intel)	6.80 0x80003d17 18.8.9	i40e 2.3.2-k (default with iso)	■ an-v1.0-osp13-rhel-server-7.7-x86_64.iso ■ rhosp13_04.tgz
Dell R740	OpenStack Ocata	Ubuntu 16.04	■ SR-IOV ■ OVS	Fortville 25G (Intel)	6.00 0x800034e5 18.3.6	i40e 2.7.12	N/A
HP DL380 Gen 10/ Gen 9	OpenStack Queens	RedHat 7.7	■ SR-IOV ■ OVS	Mellanox 25G	fw-ConnectX4Lx-rel-14_23_8036-817753-B21_Ax-UEFI-14.16.19-FlexBoot-3.5.505.bin	mlx5_core 5.0-0 (default with iso)	■ an-v1.0-osp13-rhel-server-7.7-x86_64.iso ■ rhosp13_04.tgz
HP DL380 Gen 10/ Gen 9	OpenStack Queens	RedHat 7.7	■ SR-IOV ■ OVS	Fortville 10G (Intel)	6.80 0x80003d17 18.8.9	i40e 2.3.2-k (default with iso)	■ an-v1.0-osp13-rhel-server-7.7-x86_64.iso ■ rhosp13_04.tgz
HP DL380 Gen 9	OpenStack Queens	RedHat 7.7	■ SR-IOV ■ OVS	Niantic 10G (Intel)	0x80000886, 1.1752.0	ixgbe 5.1.0-k-rh7.6	■ an-v1.0-osp13-rhel-server-7.7-x86_64.iso ■ rhosp13_04.tgz
HP c7000 Gen 9	OpenStack Queens	RedHat 7.7	■ SR-IOV ■ OVS	Niantic 10G (Intel)	0x80000886, 1.1752.0	ixgbe 5.1.0-k-rh7.6	■ an-v1.0-osp13-rhel-server-7.7-x86_64.iso ■ rhosp13_04.tgz

^a The supported switching technology applies to all interfaces.

Note: For existing Newton installations, upgrading to this MCC release is supported.

HPE Service Pack for ProLiant Versions

The following Hewlett Packard Enterprise (HPE) Service Pack for ProLiant (SPP) versions and hardware platforms are qualified by Affirmed Networks for use with this release.

HPE SPP Version	Hardware Platform
2019.03.1	■ HP c7000 Gen 9 ■ HP DL380 Gen 10

Minimum Host Server and Disk Requirements

Affirmed Networks recommends the following minimum host server and disk requirements:

- Server must have a hardware-based storage controller.
- Disks must meet the following minimum requirements:
 - Serial Attached SCSI (SAS) interface
 - 10K RPM hard disk drive
 - Disk buffer

HPE Server BIOS Settings

The following table lists the required Affirmed Networks BIOS settings for HPE servers. All other BIOS parameters do not affect the operation of Affirmed Networks software.

Parameter	BIOS Setting
HPE Gen8/Gen9 Servers	
Hyperthreading	Enabled
CPU_Performance	IO
HP_Power_Regulator	HP_Static_High_Performance_Mode
CPU_Virtualization	Enabled
Low_Power_Halt_state	Disabled
Intel_Processor_Core_Disable	All_Cores_Enabled
Intel_Processor_Turbo_Mode	Enabled
Intel_VT-d2	Enabled

Parameter	BIOS Setting
HP_Power_Profile	Maximum_Performance
Intel_QPI_Link_Power_Management	Disabled
Intel_Minimum_Processor_Idle_Power_State	No_C-State
Intel_Minimum_Processor_Idle_Power_Package_State	No_Package_State
Thermal_Configuration	Increased_Cooling
Intel_Hyperthreading	Enabled
Intel_Turbo_Boost_Optimization	Optimized_for_Performance
Intel_Turbo_Boost_Optimization_Gen8	Enabled
Energy_Performance_Bias	Maximum_Performance
Energy_Performance_Bias_Gen9	Maximum_Performance
Intel_DIMM_Voltage_Preference	Optimized_for_Performance
SR-IOV	Enabled
HPE Gen10 Servers	
Workload_Profile	Custom
Dynamic_Power_Capping_Functionality	Disabled
Intel_Hyper-Threading	Enabled
Enabled_Cores_per_Processor	"00"
Processor_x2APIC_Support	Enabled
Maximum_Memory_Bus_Frequency	Auto
Intel_VT-d	Enabled
SR-IOV	Enabled
Boot_Mode	UEFI_Mode
UEFI_Optimized_Boot	Enabled
Power_Regulator	Static_High_Performance_Mode
Minimum_Processor_Idle_Power_Core_C-State	No_C-states
Minimum_Processor_Idle_Power_Package_C-State	No_Package_State
Intel_Turbo_Boost_Technology	Enabled
Energy/Performance_Bias	Maximum_Performance
Collaborative_Power_Control	Disabled

Parameter	BIOS Setting
NUMA_Group_Size_Optimization	Clustered
Sub-NUMA_Clustering	Disabled
Energy_Efficient_Turbo	Disabled
Intel_UPI_Link_Power_Management	Disabled
Thermal_Configuration	Increased_Cooling
Thermal_Shutdown	Disabled

VM Resource Requirements

The following tables list the required resources for each SR-IOV VM and non-SR-IOV VM type.

Table 4 VM Resource Requirements for SR-IOV

VM	GB Memory (minimum / suggested) ¹	GB Memory (maximum qualified)	vCPUs (minimum / suggested) ¹	vCPUs (maximum qualified)	GB Disk Size Required for Image Deployment (main disk)	GB Disk Size Required for Any Other Disk on the Node	Total GB Disk Size KVM ²	Total GB Disk Size ESXi ³
VM Resource Requirements for SR-IOV								
ASM	32 / 110	110	16 / 22	22	50	50 when using content categorization or MMS services, 500 when using content caching. Secondary disk only required if deploying content cache, content categorization, or MMS.	550	582
CCM	32 / 56	32	8 / 16	16	50	—	50	82
CPM	32 / 56	240	8 / 16	20	50	—	50	82
CSM ⁴	56 / 124	56	8 / 16	16	50	500 (Affirmed Cache Disk) Required only if deploying content cache, content categorization, or MMS	550	606
DCM	32 / 56	110	8 / 16	16	50	—	50	82
EMS	32 / 32	64	16 / 16	—	200	—	200	232
IOM	32 / 62	110	16 / 20	28	50	—	50	82
ISM	64 / 128	128	16 / 40	40	50	50 when using content categorization or MMS services, 500 when using content caching. Secondary disk only required if deploying content cache, content categorization, or MMS.	550	614
MCM	16 / 32	32	8 / 16	16	112	400 (Affirmed Extended Storage Partition)	512	528
SSM	32 / 62 ⁵	128	16 / 20	40	50	—	50	82
WSM	32 / 62	110	16 / 20	20	50	—	50	82

¹ Minimum value required for system boot up and stability. Suggested value for deployment. Note: MCC service levels will be affected by the available hardware resources.

² This value is determined by adding the GB disk size for image + GB disk size for other disk.

³ This value is determined by adding the minimum GB memory + GB disk size for image + GB disk size for other disk.

⁴ This VM should only be used for a small POC.

⁵ 124 GB is required for ePDG.

Table 5 VM Resource Requirements for Non-SR-IOV

VM	GB Memory (minimum / suggested) ¹	GB Memory (maximum qualified)	vCPUs (minimum / suggested) ¹	vCPUs (maximum qualified)	GB Disk Size Required for Image Deployment (main disk)	GB Disk Size Required for Any Other Disk on the Node	Total GB Disk Size KVM ²	Total GB Disk Size ESXi ³
VM Resource Requirements for Non-SR-IOV								
ASM	32 / 56	—	8 / 8	—	50	500 (Affirmed Cache Disk) Required only if deploying content cache, content categorization, or MMS	550	582
CCM	32 / 56	—	8 / 16	—	50	—	50	82
CPM	32 / 56	—	8 / 16	—	50	—	50	82
CSM ⁴	56 / 124	56	8 / 12	12	50	500 (Affirmed Cache Disk) Required only if deploying content cache, content categorization, or MMS	550	606
DCM	32 / 56	—	8 / 16	—	50	—	50	82
EMS	32 / 32	64	16 / 16	—	200	—	200	232
IOM	32 / 62	—	8 / 8	—	50	—	50	82
ISM	64 / 128	—	12 / 24	—	50	50 when using content categorization or MMS services, 500 when using content caching. Secondary disk only required if deploying content cache, content categorization, or MMS.	550	614
MCM	16 / 32	32	8 / 8	8	112	400 (Affirmed Extended Storage Partition)	512	528
SSM	32 / 62 ⁵	56	8 / 8	12	50	—	50	82
WSM	32 / 62	—	8 / 8	—	50	—	50	82

¹ Minimum value required for system boot up and stability. Suggested value for deployment. Note: MCC service levels will be affected by the available hardware resources.

² This value is determined by adding the GB disk size for image + GB disk size for other disk.

³ This value is determined by adding the minimum GB memory + GB disk size for image + GB disk size for other disk.

⁴ This VM should only be used for a small POC.

⁵ 124 GB is required for ePDG.

Performing an Upgrade

See the *Affirmed Networks System Administration Guide* for details on preparing for the upgrade and performing the upgrade. **Contact the Affirmed Networks TAC for assistance with the upgrade procedure.**

CAUTION: Performing an upgrade to an older release is not supported. For example, you cannot upgrade from release 9.4 to 8.4. Instead, use **image-management rollback** to revert to the release/configuration prior to upgrading.

This release uses additional CPU cores for packet processing which must be isolated from Linux to prevent it from scheduling additional processes on the cores. This requires a change to the kernel's boot command line parameter called *isolcpus* which is applied only when the nodes are rebooted. After an upgrade, the affected nodes generate the *anIsolCpusMismatch* alarm to indicate they must be rebooted. Use the **graceful-restart** CLI command to reboot all affected nodes and complete the upgrade. See the *Affirmed Networks SNMP MIB and Alarm Reference Guide* for additional information.

Upgrade Sequence

Note: To expand the cluster from 32 VMs to 128 VMs, you must first install the latest OVAs. See [Restrictions and Special Considerations \(page 55\)](#).

Upgrade the MCC components in the following order:

- 1 Upgrade Acuitas. For instructions, see the *Software Release Notes for the Acuitas Service Management System*.
- 2 Upgrade MCC to your specific release.
- 3 Upgrade vCenter to 6.5.
- 4 Add the ESXi hosts that make up the MCC over to the 6.5 vCenter server.
- 5 Upgrade ESXi to ESXi 6.5 Update 2. Contact the TAC for instructions.
- 6 Convert Hard Disk to SCSI Controller and Network Adapters to VMXNET3. Contact the TAC for instructions. Note: This conversion applies to an upgrade from ESXi 5.5.
- 7 Use the **show alarms active** command to monitor active alarms. If necessary, perform any corrective actions specified by the alarms to complete the upgrade. See [How to Obtain Product Documentation \(page 4\)](#).

- 8 After upgrading to release 9.2, you must issue the following command on each UPF for the user planes to be fully functional:
`an3000-mcm-slot7cpu# enable-gw-agent`
- 9 (Optional) Use the `show cluster <id> node <id> processes | include GwAgent` command to verify that the GwAgent processes are running.

If the Upgrade Fails

In the event that the upgrade fails, enter `show image-management upgrade-progress`. If you encounter output similar to the following, it may indicate that the disk is full.

```
12:48:09.66->1-mm-7-1: error preparing tos 4.2.2.0-192.REL9.4
```

Online Software Upgrade from Release 9.2.x/9.3.x to 9.4.x for CUPS

Use the following steps to perform an online software upgrade (OSU) to release 9.4.x on redundant CUPS clusters with one CPF and one or more UPFs running release 9.2.x/9.3.x.

- 1 On a single UPF, run OSU from 9.2.x/9.3.x to 9.4.x.
- 2 Use the `show cluster summary` command to display the operational status of the nodes and ensure that they are all operationally up before proceeding to [step 3](#).
- 3 Repeat [step 1](#) through [step 2](#) until all UPFs are upgraded and operationally up, then proceed to [step 4](#).
- 4 On the CPF, run OSU from 9.2.x/9.3.x to 9.4.x.

Re-OVA Requirements

The following table shows the re-OVA requirements for upgrades.

Release	5.x	6.x	7.x	8.x	9.x
5.x	No	No	No	No	No
6.x	N/A	No	No	No	No
7.x	N/A	N/A	No	No	No
8.x	N/A	N/A	N/A	No	No
9.x	N/A	N/A	N/A	N/A	No

Supported Upgrade Paths

This section shows the supported upgrade paths to Release 9.4.0.0 for the VM platforms.

Figure 1 Supported Upgrade Paths

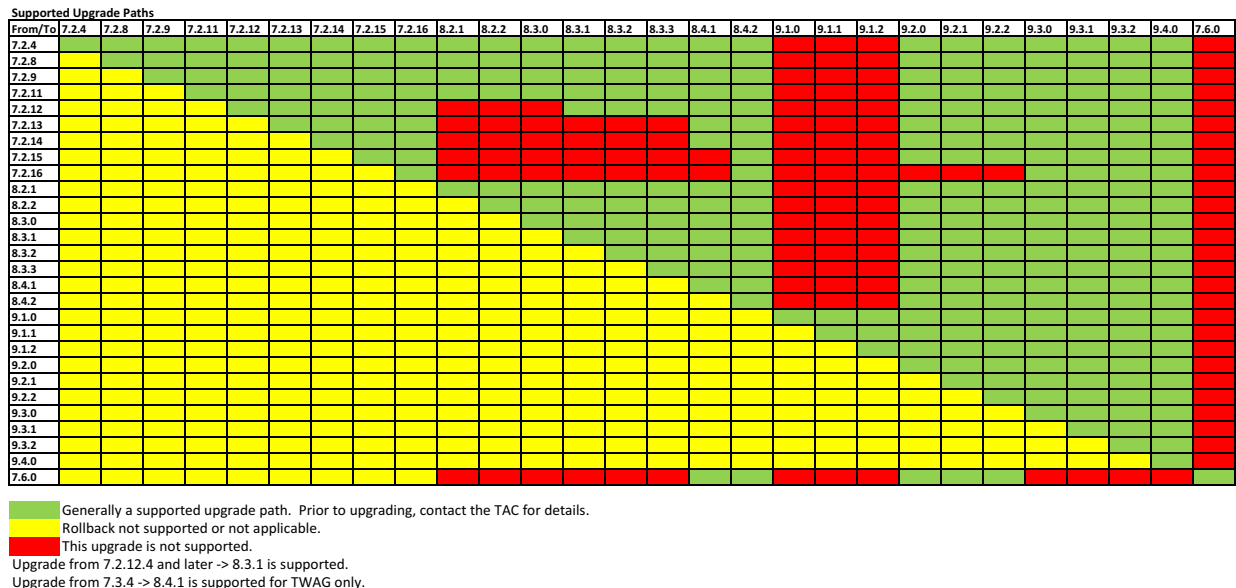
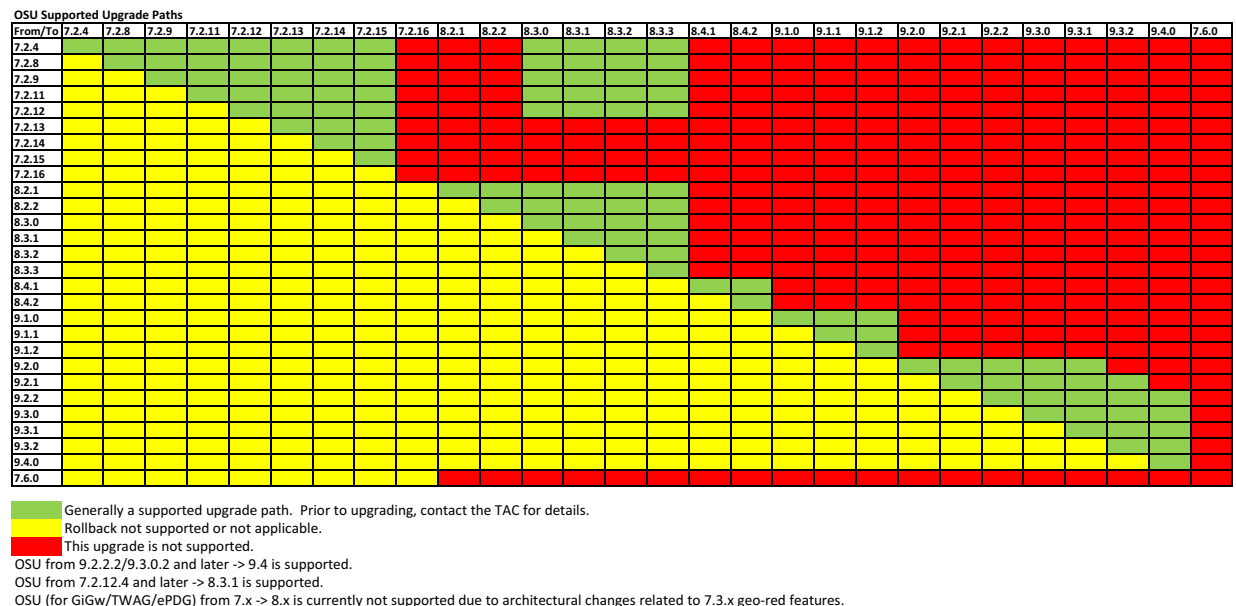


Figure 2 OSU Supported Upgrade Paths



Prerequisite Upgrade Tasks

Prior to performing an upgrade, verify or complete the following:

Note: See [Restrictions and Special Considerations \(page 55\)](#) and [Unsupported Features \(page 45\)](#) for additional details that may pertain to an upgrade.

- Check the disk space.
- See [Configuring IPv4 Address Families in a BGP Configuration \(page 18\)](#).

Configuring IPv4 Address Families in a BGP Configuration

If you are using BGP, the IPv4 address family is not configured automatically on each neighbor. The IPv4 address family is configured in a manner similar to other address families. For each neighbor for which you want to have an IPv4 address family configured, you must explicitly add the neighbor to the IPv4 address family.

For example, under `ip-protocols bgp` or `bgp-vpn`:

```
ipv4-ucast-af
  neighbor NYC_1
  neighbor NYC_2
ipv6-ucast-af
  neighbor NYC_v6
neighbor NYC_1
  neighbor-addr 10.10.1.2
  remote-as 200
  timers keep-alive 30
  timers hold-time 90
!
neighbor NYC_2
  neighbor-addr 10.10.2.2
  remote-as 200
  timers keep-alive 30
  timers hold-time 90
!
neighbor NYC_v6
  neighbor-addr 3000:1:1:2::2
  remote-as 200
  timers keep-alive 30
  timers hold-time 90
```

In this example configuration, neighbors NYC_1 and NYC_2 have a configured IPv4 address family but no configured IPv6 address family. NYC_v6 has a configured IPv6 address family but no configured IPv4 address family.

New Features and Enhancements

This release includes features from the 9.3.2.3, 9.3.1.2, 9.3.0.2, 9.2.2.11, 8.4.2.6, 8.4.1.16, 8.3.3.8, 8.3.2.5, 7.9.4.11, 7.2.16.9, and 7.2.13.6 releases. See the applicable release notes for details.

Release 9.4.0.0 includes the following new features or enhancements. For information on statistics associated with new features, see the applicable release of *Software Release Notes for the Acuitas Service Management System* as referenced in the [Release Abstract \(page 4\)](#). For information on limitations associated with new features, see [Unsupported Features \(page 45\)](#).

Feature	Description
Support for documentation listing the maximum number of supported Mobile Content Cloud object instances, by major MCC release	Documentation is available listing the maximum number of supported Mobile Content Cloud object instances, by major MCC release. See Mobile Content Cloud Maximum Supported Object Instances (page 83) .
Added support for the Sd interface between the PCRF and the GiGW (acting as TDF) See the <i>Affirmed Networks Gateway Administration Guide</i> .	The Sd interface is located between the PCRF and the Traffic Detection Function (TDF) and is used to detect traffic and report information on the detected application traffic. This feature adds support for the Sd interface between the PCRF and the GiGW (acting as a TDF) as defined in 3GPP 29.212, Release 15.3.0. The GiGW has a new Diameter session-learner interface used to establish and terminate TDF sessions to the PCRF.

Feature	Description
Added ability to provide variable QoS for traffic based on information retrieved from a manifest server and traffic detection using ADC over the Gx interface See the <i>Affirmed Networks Gx Interface Specification</i>.	This feature adds the ability to provide variable QoS for traffic based on information retrieved from a manifest server and traffic detection using ADC over the Gx interface. This feature adds support for the acwsponsorurl vendor-specific attribute (VSA) on the Gx interface. If the PCRF sends the acwsponsorurl VSA along with the Charging-Rule-Definition, TDF-Application-Identifier and Event-Trigger APPLICATION_START and APPLICATION_STOP in the CCA/RAR message, the MCC gateway will retrieve flow information from a manifest server and deliver this information to the PCRF via a CCR-U message. This feature adds a new parameter to configure a manifest interface profile. <pre>an3000-mcm-slot7cpu1(config)# service-construct manifest-interface-profile <manifest-interface-profile-name> admin-state <enabled/disabled> certificate-server ca-certificate-list <ca-certificate-list name> certificate-server certificate-url <cert server url/ip> certificate-server certificate-file-extension <1 to 255 char string> certificate-server network-context <network-context> certificate-server loopback-ip <local IP loopback> certificate-server crl-url <crl server url/ip> manifest manifest-max-size <5 to 512000> manifest manifest-max-size-unit <bytes/kbytes> certificate-verification admin-state <enabled/disabled></pre> The new manifest interface profile must be associated with a Workflow Control Profile. <pre>an3000-mcm-slot7cpu1(config)# workflow control-profile <control-profile-name> manifest-interface-profile <manifest-interface-profile-name></pre> This feature adds support for processing the Custom-Mute VSA received in CCA/RAR messages on the Gx interface. The Custom-Mute VSA is used to limit APPLICATION_START/STOP notifications sent to the PCRF. This feature also adds support for sending the Sponsored-Connectivity-Data AVP in the CCR-U message on the Gx interface. If the manifest response comes with a validity time, the MCC includes the validity time in the CC-Time AVP within the sponsored connectivity data and starts the validity timer for the manifest refresh.
Support for Dell R740 running ESXi 6.5 U2 with Fortville 25G NICs	This release adds support for the Dell R740 running VMware ESXi version 6.5 U2 with Fortville 25G NICs on MCC SR-IOV systems.
Support for Dell R740 running ESXi 6.7 U2 with Fortville 25G NICs	This release adds support for the Dell R740 running VMware ESXi version 6.7 U2 with Fortville 25G NICs on MCC SR-IOV systems.

Feature	Description
Added user account password security improvements. See the <i>Affirmed Networks System Administration Guide</i>.	Added user account password security improvements in this release. Use the following command to configure password policies: <pre>an3000-mcm-slot7cpu1# user-access password-policies</pre> Use the following command to display the current password policies by domain: <pre>an3000-mcm-slot7cpu1# show user-access password-policies default domain policy Numeric Complexity : 0 range 0 - 6 Uppercase Complexity : 1 range 0 - 6 Lowercase Complexity : 1 range 0 - 6 Non-alphanumeric Complexity : 1 range 0 - 6 Limit on sequential characters : 0 range 0 - 64 Limit on repeated characters : 0 range 0 - 64 Allow username in password : 1 range 0 - 1 [0=no 1=yes] Minimum Password Length : 8 range 0 - 32 Password Dictionary : 0 range 0 - 2 [0=no 1=default 2=custom] Previous Password History : 3 range 0 - 30 Lockout Failure Limit : 5 range 0 - 6 Lockout Duration : 10 range 0 - 60 [minutes] Minimum Password Age : 0 range 0 - 7 [day(s)] Maximum Password Age : 0 range 0 - 365 [day(s)] admin domain policy (unset values use defaults)</pre>
Allow Martian routes to be advertised by BGP	Martian addresses are addresses reserved by IANA for special use. The following addresses are reserved for private networks and treated as Martian routes. 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, 169.254.0.0/16 This feature allows BGP to advertise Martian routes created for UE Pools and loopbacks. You can use route maps to suppress Martian routes.

Feature	Description
Enhanced tethering detection logic using GQUIC user agent fingerprinting	Enhanced tethering detection logic using GQUIC user agent fingerprinting. Added the following parameter to enable running the tethering detection algorithm using GQUIC user agent fingerprinting: <pre>an3000-mcm-slot7cpu1(config)# data-path dpi tethering-detection-algorithm gquic-user-agent <enable disable></pre> By default, this parameter is disabled. Changes to this setting take effect upon next reboot. Added the following parameter to configure the weight percentage for the GQUIC user agent quality index: <pre>an3000-mcm-slot7cpu1(config)# data-path dpi tethering-quality-index-result-weight percentage gquic-user-agent <percentage></pre> Added the following statistics to indicate the number of tethered flows detected by GQUIC user agent fingerprinting: <pre>an3000-mcm-slot7cpu1# show subscriber pdn-session <session-id> an3000-mcm-slot7cpu1# show zone default deep-packet-inspection statistics grid-level tethering <id> an3000-mcm-slot7cpu1# show zone default gateway statistics [cluster-level grid-level node-level] call-performance apn <id> an3000-mcm-slot7cpu1# show zone default gateway statistics [cluster-level grid-level node-level] call-performance pgw <id> tethered-flows-detected-using-gquic-user-agent</pre> In the FLOW EDR, added the GQUIC user agent value to the tetheringAlgorithm field: tetheringAlgorithm <string> {ttl, userAgent, tcpipfingerprinting, gquic-user-agent}
Proxy is TTL transparent See the <i>Affirmed Networks Value Added Services Guide</i> .	Previously, for upstream traffic from the UE to the origin server, IP packets that <i>traversed</i> the proxy had their TTL values reset to 63 for IPv4, and their hop limit values reset to 254 for IPv6. However, IP packets that <i>bypassed</i> the proxy had their TTL and hop limit values correctly decremented by one. With this feature, traffic traversing the proxy also has the TTL/hop limit value decremented. Specifically, for any HTTP, HTTPS, and QUIC flows originating from the UE for IPv4 and IPv6, the TTL or hop limit value is decremented by one, respectively. This applies to transparent IP and non-transparent IP traffic traversing the proxy, and to traffic bypassing the proxy, regardless of whether the User Space IP stack or the Linux IP stack is used. Note: OSU – During an upgrade, traffic traversing VM nodes that have not been upgraded will retain the existing behavior.

Feature	Description
Support for IP address filters See the <i>Affirmed Networks Gateway Administration Guide</i> .	This feature adds support for the ipv4-filter-list object, which filters IPv4 packets by matching one or more ranges of IP addresses. Prior to this release, you had to use the packet-filter object to filter packets by IP address ranges. The ipv4-filter-list is more efficient than the packet-filter when you are filtering only on IP addresses and a service-rule would require hundreds of packet-filters. You can configure 10 ipv4-filter-lists. An ipv4-filter-list supports IPv4. IPv6 is <i>not</i> currently supported. To configure an ipv4-filter-list: <pre>an3000-mcm-slot7cpu1(config)# service-construct ipv4-filter-list <name> filter-type [network ue] <CIDR range></pre> where: ■ filter-type [network ue] — Specifies whether the filter is for a network initiated connection or for a user equipment initiated connection. ■ <CIDR range> — Specifies one or more CIDR ranges for the IP addresses, for example, 3.1.1.0/24. You can enter 4000 CIDR ranges for each ipv4-filter-list. To include the ipv4-filter-list in a service-rule: <pre>an3000-mcm-slot7cpu1(config)# service-construct service-rule <name> ip-filter-list <name></pre>
Support for service-rule logical OR for matching conditions See the <i>Affirmed Networks Gateway Administration Guide</i> .	This feature adds the new evaluation-mode parameter to specify how many of the configured classifier types must be matched for a service-rule to be considered a match: <pre>an3000-mcm-slot7cpu1(config)# service-construct service-rule <name> evaluation-mode [AND OR]</pre> where: ■ AND – Each type of the configured classifiers must be matched. ■ OR – At least one type of the configured classifiers must be matched. Prior to this release, only the AND behavior existed.
Enhancements for flow-idle-timeout-filter See the <i>Affirmed Networks Gateway Administration Guide</i> .	The flow-idle-timeout-filter now supports the following configuration options: source IP address range, destination IP address range, and IP type. These new options are in addition to the existing destination port range and IP protocol options. To configure the new options: <pre>an3000-mcm-slot7cpu1(config)# service-construct flow-idle-timeout-filter <name> ip-version [v4-only v4-or-v6 v6-only] network-ip-address-end <address> network-ip-address-start <address> network-ip-v6-address-end <address> network-ip-v6-address-start <address> ue-ip-address-end <address> ue-ip-address-start <address> ue-ip-v6-address-end <address> ue-ip-v6-address-start <address></pre>

Feature	Description
Added support for modifying QCI/ARP values of the EPS dedicated bearer if the PCRF uses the Charging-Rule-Install AVP to modify QCI/ARP values See the <i>Affirmed Networks Gx Interface Specification</i>.	Each unique QoS Class Identifiers/Allocation and Retention Priority (QCI/ARP) combination identifies a bearer on the PGW. This feature adds support for the following two scenarios if the PCRF uses the Charging-Rule-Install AVP to modify QCI/ARP values. ■ If there are other charging rules on the existing dedicated bearer, the PGW keeps the existing dedicated bearer but removes the particular charging rule for which QCI/ARP values have been modified. The PGW creates a new dedicated bearer with the new QCI/ARP values to carry the traffic for this charging rule. ■ If there is already an existing dedicated bearer with the same QCI/ARP values as those in the charging rules being modified, the PGW moves those modified charging rules to the existing bearer and deletes the old dedicated bearer if no charging rules are left on that bearer.

Feature	Description
Added support for throttling control plane GTP messages on the PGW and SGW when the system is in an overload state See the <i>Affirmed Networks Gateway Administration Guide</i>.	Added support for throttling the control plane GTP messages on the PGW and SGW when the system is in an overload state. Throttling is based on configurable ARP priority levels. Emergency call setups are not affected by this feature. To configure throttling, use the new overload-control max-arp-priority-level command: <pre>an3000-mcm-slot7cpu1(config)# zone default gateway profile gateway-common overload-control max-arp-priority-level minor <priority> overload-control max-arp-priority-level major <priority> overload-control max-arp-priority-level critical <priority></pre> where <priority> is a bearer's ARP priority level. When the system is in an overload state, the thresholds are used as follows: ■ Create session requests are rejected if the default bearer ARP priority level is greater than the max-arp-priority-level. ■ Create bearer requests and bearer resource command messages are rejected if the session does not have any bearers with an ARP priority level that is less than or equal to the max-arp-priority-level. Note: All new session setups are rejected when the system is in overload state and the MCD table sync to standby process is in progress. An overload condition can be any of the following. If more than one overload condition exists, the highest overload state is used for throttling. ■ PGW and SGW tasks exceed the CPU overload states: – minor — 80 percent – major — 88 percent – critical — 95 percent ■ The number of PGW and SGW create-session-requests exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command: <pre>an3000-mcm-slot7cpu1(config)# zone default gateway threshold node-level call-performance pdngw create-session-requests-[1 5 15]-minute-average major-enter <int> minor-enter <int> critical-enter <int> admin-state enabled</pre> ■ The node-level memory consumption percentage exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command: <pre>an3000-mcm-slot7cpu1(config)# zone default system threshold node-level memory-stats csm-avg-[1 5 15]min-memory-util minor-enter <int> major-enter <int> critical-enter <int> admin-state enabled</pre>

Feature	Description
Added support for throttling control plane GTP messages on the PGW and SGW when the system is in an overload state (continued)	■ The node-level CPU consumption percentage exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command: <pre>an3000-mcm-slot7cpu1(config)# zone default system threshold node-level cpu-stats csm-avg-[1 5 15]min-cpu-util minor-enter <int> major-enter <int> critical-enter <int> admin-state enabled</pre> The following command displays the new overload statistics: <pre>an3000-mcm-slot7cpu1# show zone default gateway statistics cluster-level call-performance [pgw sgw] include overload</pre> <pre>num-create-session-rejects-overload num-create-bearer-rejects-overload</pre> The following alarms are generated when the system is in an overload state: <pre>anBladeOverload anThresholdCrossingAlarm</pre>
Updated DPI script to validate user-defined signatures and use image management techniques to load the packages dynamically See the <i>Affirmed Networks DPI Script User Guide</i>.	The DPI script enables a Network Operator to add or update classification support for applications and protocols without requiring an MCC software update or restart. The DPI script validates user-defined signatures and increases system performance by using image management techniques to load the packages dynamically. Configure the dpiscript parameters as follows: <pre>an3000-mcm-slot7cpu1(config)# data-path dpi dpiscript</pre> ■ script-classification – Set to enabled or disabled (default). In the disabled state, user-defined signatures are loaded on the system but the classification/detection phase does not consider these signatures. ■ http-classification – Set to enabled or disabled for dpiscript http support. This parameter takes effect only when script-classification is enabled. ■ https-classification – Set to enabled or disabled for dpiscript https support. This parameter takes effect only when script-classification is enabled. See the <i>Affirmed Networks DPI Script User Guide</i> for information on configuring and creating a DPI script and collecting and analyzing pcaps.
Support for configurable Tethering Quality Index (QI)	In previous MCC releases, the Quality Index that rates the confidence level of DPI-detected tethering used hard coded values. This feature supports configurable values. A connection is marked as tethered only if the reported Quality Index (Confidence Index) is greater than or equal to the detection algorithm QI value configured using the following command: <pre>data-path dpi tethering-quality-index value <QI> [default = 70]</pre>
Added TTL value to the dpiStringInfo subfield of the FlowRecord field of the Flow EDR	Added the TTL value to the dpiStringInfo subfield of the FlowRecord field of the Flow EDR.

Feature	Description
Added support to encrypt non-transient in-memory and inter-VM identities for Lawful Intercept (LI) targets	Added support to use AES-256 to encrypt non-transient in-memory and inter-VM identities for Lawful Intercept (LI) targets, including: ■ Identities: IMSI, IMEI, MSISDN ■ Lawful Intercept Identifier (LIID) ■ Correlation ID The LI Administrator (ADMF) configures the AES-256 encryption key using the Public Key Infrastructure (PKI) CLI: <pre>pki install type calea name target-key encrypt object <key-text></pre> <pre>pki remove type calea name target-key</pre> where <key-text> is always 32 bytes and can be specified in one of the following formats: ■ raw bytes – 32 characters ■ base64 – 44 characters ■ hex – 64 characters Issue the following command to verify that AES-256 encryption is enabled: <pre>show calea service-status</pre> The command displays the encryption statistics, including: <pre>target-encryption-type aes256</pre> which indicates that the encryption is enabled.
Added support to use AES-256 to encrypt all Lawful Intercept (LI) information passed using PFCP over Sxa and Sxbc interfaces	Added support to use AES-256 to encrypt all LI information passed using PFCP over Sxa and Sxbc interfaces. These interfaces apply only to CUPS configurations. Previous Sxa and Sxbc interface encryption supported only MD5. The LI Administrator (ADMF) enables encryption: <pre>calea pfcplawfulinterceptie adminstate enabled encryption [disabled md5 aes256]</pre> (Default: disabled) In addition: ■ To enhance security, configure a secret-key: <pre>calea pfcplawfulinterceptie secretkey <key-text></pre> ■ If the secret-key <i>is not</i> configured, but an LI Target Encryption is configured, the specified AES-256 key is used as the secret-key: <pre>pki install type calea name target-key encrypt object <key-text></pre> ■ If neither is configured, a default hard-coded key is used.

Feature	Description
Added support for populating the UWAN User Location Information field in local, GTPP and Gz/Rf PGW CDRs when the UE Local IP Address and the UE UDP Port IE are received from the ePDG See the <i>Affirmed Networks Charging Data Records Specification</i>.	When the UE Local IP Address and the UE UDP Port are received from the ePDG, the MCC populates the UWAN User Location Information field in local (ASN.1 and CSV), GTPP and Gz/Rf PGW CDRs. This CDR field is configurable in local ASN.1 and GTPP Interim and Stop data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the data field is not present in CDRs. Top-level CDR field <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ofcs gtp [interim stop] <template-name> [uwanuserlocationinfo uwanuserlocationinfofoudpsourceport] presentFlag [yes no conditional]</pre> Within the Change of Charging Condition Field (List of Traffic Data Volumes) and the Change of Service Condition field (List of Service Data) <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ofcs gtp [interim stop] <template-name> [changeofcharconditionuwanuserlocationinfo changeofcharco nditionuwanuserlocationinfofoudpsourceport changeofservicec onditionuwanuserlocationinfo changeofserviceconditionuwan userlocationinfofoudpsourceport] presentFlag [yes no conditional]</pre> This CDR field is configurable in local CSV Start, Interim and Stop data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the data field is not present in CDRs. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw charging-local [start interim stop event] <template-name> uwanuserlocationinfo presentFlag [yes no conditional]</pre> The UWAN-User-Location-Information AVP is configurable in ACR data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the field is not sent. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ofcs diameter [start interim stop event] <template-name> uwanuserlocationinfo presentFlag [yes no conditional]</pre>

Feature	Description
Support for Multi-Access Edge Computing local breakout for CUPS See the <i>Affirmed Networks Gateway Administration Guide</i> .	Added support for Multi-Access Edge Computing (MEC) local breakout on a standalone CUPS SGW. Local breakout allows traffic that only needs local applications to be serviced at the operator edge site. This reduces latency for local applications, and reduces back-haul requirements. Configure local breakout on a CUPS standalone SGW CPF: Enable local breakout in a user-plane-profile. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway user-plane-profile <name> mec-local-breakout enable</pre> Configure a user-plane-selector type of subscriber-analyzer that specifies the user-plane-profile that enables local breakout. <pre>an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer <name> analyzer-type user-plane-selector priority <priority> action user-plane-profile <name> key <list of keys></pre> When a key in this subscriber-analyzer is matched, the data-profile from the matching workflow-profile type of subscriber-analyzer is sent to the UPF. Data packets matching the data-profile undergo a local breakout.
Enhancements for ip-address-allocation-mode See the <i>Affirmed Networks Gateway Administration Guide</i> .	The ip-address-allocation-mode now supports the new options ipv4v6-prefer-v4 and ipv4v6-prefer-v6. The ip-address-allocation-mode specifies how the type of allocated address (IPv4 or IPv6) is chosen when the UE requests IPV4IPV6. To configure the new options: <pre>an3000-mcm-slot7cpu1(config)# zone default gateway apn <name> ip-address-allocation-mode [ipv4 ipv6 ipv4v6 ipv4v6-prefer-v4 ipv4v6-prefer-v6]</pre> New options: ■ ipv4v6-prefer-v4 – Allocate an IPv4 address if it is available; otherwise allocate an IPv6 address if it is available. If neither type of address is available, the session fails. ■ ipv4v6-prefer-v6 – Allocate an IPv6 address if it is available; otherwise allocate an IPv4 address if it is available. If neither type of address is available, the session fails. Options that already existed: ■ ipv4 – Allocate an IPv4 address if one is available. If an IPv4 address is not available, the session fails. ■ ipv6 – Allocate an IPv6 address if one is available. If an IPv6 address is not available, the session fails. ■ ipv4v6 – If a dual address bearer flag (DAF) is received from the UE, allocate both an IPv4 and an IPv6 address. If a DAF is not received, allocate an IPv4 address if it is available; otherwise allocate an IPv6 address if it is available. If neither type of address is available, the session fails. Note: If the UE requests IPv4 or requests IPv6, that type of address is allocated if it is available, regardless of the ip-address-allocation-mode. If the requested type of IP address is not available, the session fails.
Added support for applications and protocols supported by the DPI Signature in MCC R9.4 and later releases	Added support for applications and protocols supported by the DPI Signature in MCC R9.4 and later releases. For details, see <i>Affirmed Networks DPI Classification Release Notes for R8.4 and Above</i>.

Feature	Description
Added support for the CGI-SAI change and ECGI change event-triggers and increased the maximum number of SDCs that the gateway maintains per CDR from 20 to 50 See the <i>Affirmed Networks Charging Data Records Specification</i> and the <i>Affirmed Networks Gy Interface Specification</i>.	This feature adds two new event-triggers to the <code>event-trigger-template</code>, which is used to specify the events that trigger an interim update on offline/online charging interfaces and to trigger the closure of an interim SGW/PGW/GGSN CDR. <i>Applies to the Gy, Gz/Rf and Ga (GTPP), local (ASN.1) interfaces.</i> For all event-triggers in the <code>event-trigger-template</code>, the maximum service data container (SDC) limit per CDR closure event is increased from 20 to 50. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface-trigger-template event-trigger-template <template-name> [cgi-sai-change-close-trigger ecgi-change-close-trigger]</pre> ■ cgi-sai-change-close-trigger – Specifies whether a change in the Cell Global Identification- Service Area Identifier (CGI-SAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CGI-SAI change events to trigger a CDR close. ■ ecgi-change-close-trigger – Specifies whether a change in the E-UTRAN Cell Global Identifier (ECGI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of ECGI change events to trigger a CDR close. This feature also increased the maximum number of SDCs that the gateway maintains per CDR at the bearer-level from 20 to 50 if interim-updates are enabled on the interface. Upon reaching this limit, the gateway transfers the SDCs in an interim CDR to the offline server. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [pgw sgw ggsn wag epdg] num-service-container [1..50]</pre> Added the following change in charging conditions to the Change-Condition AVP within the Service-Information → PS-Information AVP sent on the Gz/Rf interface. 14 CGI-SAI Change 16 ECGI Change Added the following change in charging conditions to the Change-Condition ASN.1 field within the Change of Service Condition field sent on the Ga interface. cGI-SAICChange(21) eCGICChange(29) Added the following change in charging conditions to the Change-Condition ASN.1 field within the Change of Charging Condition field sent on the Ga interface. cGI-SAICChange(6) eCGICChange(10)

Feature	Description
Added support for the CGI-SAI change and ECGI change event-triggers and increased the maximum number of SDCs that the gateway maintains per CDR from 20 to 50 (continued)	Added two new configuration parameters under the offline-charging interface: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [pgw sgw ggsn wag epdg] num-charge-condition-change [0..50]</pre> ■ num-charge-condition-change – Specifies the maximum number of change in charging conditions (such as RAT change, QoS change, ULI change, etc.) that the gateway maintains per CDR. Upon reaching this limit, the gateway transfers the SDCs in an interim CDR to the offline server. The default is 0. <i>Note: A trigger is only counted towards the num-charge-condition-change if it is enabled under the event-trigger-template.</i> <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [pgw ggsn] qos-change-trigger [pcrf-initiated access-nw-initiated any]</pre> ■ qos-change-trigger – Specifies when to activate the qos-change-trigger. – pcrf-initiated – Activate the qos-change-trigger for a PCRF-initiated QoS change only. – access-nw-initiated – Activate the qos-change-trigger for an access-network/UE-initiated QoS change only. – any – (default) Activate the qos-change-trigger for any type of QoS change.
Added support for the CAMEL Charging Information CDR field and added 3GPP 32.298 version 10.0 as an option for formatting local ASN.1 and GTPP CDRs See the <i>Affirmed Networks Charging Data Record Specification</i>.	This feature adds support for the CAMEL Charging Information field in PGW/GGSN local ASN.1 and GTPP CDRs. The CAMEL Charging Information CDR field is configurable in local ASN.1 and GTPP Interim and Stop data record templates. The default presentFlag setting is no, indicating the data field is not present in CDRs. <i>Note: The CAMEL Charging Information CDR field is only applicable for 3G sessions.</i> <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template [pgw ggsn] ofcs gtpv [interim stop] <template-name> camelinfo presentFlag [yes no conditional]</pre> This feature also adds version 10 as an option when configuring the 3GPP 32.298 version used to format local ASN.1 and GTPP CDR fields in Interim and Stop data records. To format local ASN.1 CDRs: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [sgw pgw ggsn] local asn1-cdr-version [10.0 9.0 8.4 6.0]</pre> <i>Note: This parameter is only visible when the encode-format is set to asn1 or both.</i> To format GTPP CDRs: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface aaa-interface-group gtpv node-admin <node-name> version [10.0 9.0 8.4]</pre> Limitation When the version parameter is set to 9.0 for local ASN.1/GTPP CDRs, the Low Priority Indicator field in local ASN.1/GTPP Interim and Stop data record templates should be set to no if it is not needed. When the version parameter is set to 10.0 for local ASN.1/GTPP CDRs, the Low Priority Indicator field in local ASN.1/GTPP Interim and Stop data record templates should be set to yes (default).

Feature	Description
Added enhancements to the existing global firewall function See the <i>Affirmed Networks System Administration Guide</i>.	This feature adds the ability to provide syslog for the following scenarios: ■ Summary of count of connections dropped per session due to incoming connections exceeding the max-connections threshold configured in workflow data-profile. ■ Summary of count of packets dropped due to malformed IP/TCP header. ■ Summary of count of flows dropped due to flow rate exceeding flow-rate-limit threshold configured in security policy rule. ■ Summary of count of packets dropped due to packet rate exceeding the packet-rate-limit threshold configured in security policy rule. The following limitations apply to this feature: ■ Upon SSM crash, firewall records that were not sent to the MCM are lost. ■ Upon MCM crash, firewall records that were not sent to the syslog server are lost.
Added a new configuration option under the online-charging interface to specify either incremental or accumulative volume-based usage reporting to the OCS See the <i>Affirmed Networks Gy Interface Specification</i>.	This feature adds a new configuration option under the online-charging interface to specify whether the MCC reports the CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets (within the Used-Service-Units AVP) either accumulatively or incrementally. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> usage-reporting-mode [accumulative-mode incremental-mode]</pre> ■ accumulative-mode – The accumulated CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets is reported to the OCS in the Used-Service-Units AVP when the subscriber requests quota or when the PGW sends a CCR-U to report the final usage for the rating group. The CC-Total-Octets is reset to 0 only when the Gy session is terminated or the subscriber is cleared. ■ incremental-mode – (default) The incremental CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets is reported to the OCS in the Used-Service-Units AVP each time the subscriber requests quota. The CC-Total-Octets is reset to 0 every time the PGW reports usage to the OCS. Note: Changes to this parameter take effect immediately. If this parameter is changed mid-session, it applies to all new subscriber sessions.
Added support for a new configuration option to configure the validity-time on the Gy interface	This feature adds a new online-charging configuration option to configure a validity time that is used when the OCS does not send the Validity-Time AVP within the MSCC AVP. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> attributes validity-time <value-in-seconds></pre>
Improve DPI processing	This feature reduces MCC latency and bursty traffic pattern for DPI processing in the case of either packet drop or out of order packets.

Feature	Description
Added support for processing the Gx-TMO-Redirect-Server AVP when received from the PCRF in CCA-I/U and RAR messages See the <i>Affirmed Networks Gx Interface Specification</i>.	This feature adds support for the Gx-TMO-Redirect-Server AVP when received in the Charging-Rule-Definition AVP from the PCRF in the CCA-Initial (CCA-I), CCA-Update (CCA-U), or RAR message. When the PGW receives the Gx-TMO-Redirect-Server AVP within the Charging-Rule-Definition AVP from the PCRF, the MCC will perform a redirection for dynamic or externally-activated predefined rules. The redirection is only valid for HTTP URLs. This feature applies to online, offline, and metering rating groups. The following subfields are supported under the Gx-TMO-Redirect-Server AVP: ■ Gx-TMO-Redirect-Address-Type ■ Gx-TMO-Redirect-Server-Address ■ Gx-TMO-Append-Original-URL – If enabled, the MCC appends the original URL to the redirect URL. ■ Gx-TMO-Append-MSISDN – If enabled, the MCC appends the MSISDN to the redirect URL. ■ Gx-TMO-Append-IMSI – If enabled, the MCC appends the IMSI to the redirect URL. ■ Gx-TMO-Append-IMEI – If enabled, the MCC appends the IMEI to the redirect URL. ■ Gx-TMO-Append-MSIP – If enabled, the MCC appends the Mobil Station (MS) IP address to the redirect URL. – If both IPv4 and IPv6 MS IP addresses are allocated, the MCC appends the active IP address to the redirect URL. – For IPv6 IP addresses, the MCC uses the preferred format. ■ Gx-TMO-Deactivate-By-Redirect – If enabled, the MCC performs a one-time redirection. After the one-time redirection, the PGW will deactivate the rule. Note: If both the Redirect-Information AVP and Gx-TMO Redirect-Server AVP are received, the MCC will use the Gx-TMO Redirect-Server AVP for redirection and will ignore the Redirect-Information AVP provided the Gx-TMO-Redirect-Server AVP is enabled in the CCA data record template. The Gx-TMO-Redirect-Server AVP is configurable in Gx CCA data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the field is not processed when received in the CCA response. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw pcrf diameter cca <template-name> tmoredirectserver presentFlag [yes no]</pre> This feature also adds a new configuration parameter under the advice-of-charge object to specify whether to include the MS IP address in the redirect URL. <pre>an3000-mcm-slot7cpu1(config)# service-construct advice-of-charge <name> http include-msip [true false]</pre>

Feature	Description
Added shaping enhancement for configurable chunk sizes	This feature provides a configurable option to control the data chunk size of rate-limited packets for the HTTP, HTTPS, and UDP protocols. The data size is controlled by the time interval at which rate-limited packets are sent to the UE for a given bitrate. For example: If the configured bitrate is 1.5Mbps and the configured send-interval is 50ms (0.05sec), MCC sends the following in each interval: $\text{chunksize} = (1.5\text{Mb} / 8) * 0.05 = 9375 \text{ bytes}$ A new send-interval field is added to the abr-optimization-profile. For example: <pre>services multi-protocol-proxy abr-optimization-profile AOP admin-state enabled</pre> send-interval – Configure rate limiting send interval (in ms) (min – 40ms, max – 500ms; in multiples of 10 ms; default – 100ms) A new send-interval field is added to the traffic-control-profile. For example: <pre>services multi-protocol-proxy traffic-control-profile ACP rate-limiting</pre> send-interval – Configure rate limiting send interval (in ms) (min – 40ms, max – 500ms; in multiples of 10 ms; default – 100ms) of 10ms; default – 100ms)
Added a new configuration option under the gateway service to configure the Node ID value used in SGW/PGW/GGSN CDRs See the <i>Affirmed Networks Charging Data Records Specification</i>.	This feature adds a new configuration option used to configure the Node ID value under the gateway service, which will be used as the Node ID value in local (ASN.1 and CSV), GTPP and Gz/Rf SGW/PGW/GGSN CDRs. If this parameter is not configured, the gateway service name is used as the value for the Node ID field in SGW/PGW/GGSN CDRs. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway [sgw pgw ggsn] <gateway-service-name> node-id <node-id-string></pre> Note: The <i>node-id</i> parameter under the gateway service should not be configured or changed mid-session. If it is, the node-id in subsequent CDRs will contain the newly configured value. There is an existing configuration parameter under the offline-charging interface that allows you to specify whether the Node ID value in CDRs reports the configured gateway service name along with the Service Type ID or just the configured gateway service name. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [sgw pgw ggsn] unique-cdr-nodeid [enabled disabled]</pre> See Table 6 on page 83 for information on the MCC behavior when using the <i>node-id</i> parameter under the gateway service and the <i>unique-cdr-nodeid</i> parameter under the offline-charging interface together.
SGX Downlink Data Notification message to MME always provides IMSI See the <i>Affirmed Networks Gateway Administration Guide</i>.	SGW will always include the IMSI value in the Downlink Data Notification message if the MME provided the IMSI in the Create Session Request.

Feature	Description
Added ability to override the charging-characteristics (CC) value received in the Create Session Request message for a subscriber based on the subscriber type See the <i>Affirmed Networks Gx Interface Specification</i>.	This feature adds the ability to override the charging-characteristics (CC) value received in the Create Session Request message for a subscriber based on the subscriber type. Set parameters using the following commands. 1 Configure a charging-characteristics profile with a configured CC value for each subscriber type. <pre>an3000-mcm-slot7cpu1(config)# service-construct charging-characteristics-profile <cc-profile-name> [home-subscriber-cc-value roaming-subscriber-cc-value visiting-subscriber-cc-value] <value></pre> 2 Associate this charging-characteristics profile with a Charging-Rule-Base-Name (CRBN) profile, APN and/or gateway service. <pre>an3000-mcm-slot7cpu1(config)# services crbn-profile <crbn-profile-name> charging-characteristics-profile <cc-profile-name> an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name> charging-characteristics-profile <cc-profile-name> an3000-mcm-slot7cpu1(config)# zone default gateway pgw <pgw-service-name> charging-characteristics-profile <cc-profile-name></pre> If the Create Session Request message contains a CC value, the MCC will override this value based on the following order of precedence. The MCC also uses this order of precedence for selecting the CC value. For example, if none of the CRBN profiles have a charging-characteristics-profile associated with them, the MCC will check the 3GPP-Charging-Characteristics AVP received in the RADIUS Access-Accept message. If the RADIUS Access-Accept does not provide a CC value, the MCC will check the charging-characteristics-profile associated with the APN, etc. 1 The CC value will be selected based on the charging-characteristics-profile associated with a CRBN profile. If multiple CRBNs are received, the MCC selects the first CRBN in the CRBN list. – CRBN profile received from the PCRF – CRBN profile received from the RADIUS server – The default CRBN profile (associated with a data profile) 2 The 3GPP-Charging-Characteristics AVP received in the RADIUS Access-Accept message. 3 The charging-characteristics-profile associated with the APN. 4 The charging-characteristics-profile associated with the gateway service. 5 The CC value received in the Create Session Request/Create PDP Context Request message. 6 The default CC value configured for the Control Profile. Note: If the CC value is not configured in the charging-characteristics-profile, the MCC uses the CC value received in the Create Session Request message.

Feature	Description
Support for monitoring the different establishment states of TCP and UDP flows accepted by a subscriber firewall flow See the <i>Affirmed Networks Gateway Administration Guide</i> .	TCP and UDP flows accepted by a subscriber-firewall-flow can be monitored for different establishment states. Separate timers for each establishment state are used to remove flows and to drop unexpected TCP messages. The timers are used as follows: tcp-open-timeout — Specifies the timeout period between each message involved in the three-way TCP connection establishment. During this period, if the expected message is not received, the flow is removed. tcp-established-timeout — Specifies the timeout period for no flow activity after the three-way handshake is complete. If no flow activity is detected during this period, the flow is removed. tcp-close-timeout — Specifies the timeout period between each message involved in at the four-way TCP close operation, starting with receipt of the initial FIN from either direction. During this period, if the expected message is not received, the message is removed. udp-open-timeout — Specifies the timeout period between receiving the initial UDP packet and receiving the response UDP packet. If the response UDP packet is not received during the timeout period, the flow is removed. udp-established-timeout — Specifies the timeout period for no flow activity after the response UDP is received. If no flow activity is detected during this period, the flow is removed. In addition, a non-configurable internal timer removes a flow 30 seconds after receiving an RST. The timers can be configured in two ways: ■ Global — The same global tracking profile is used to control the timers for any subscriber-firewall-flow that enables global control. ■ Local — A subscriber-firewall-flow specifies a particular tracking profile to control timers. To globally configure monitoring of the flow establishment states: 1 Enable global control for the tracking profile selection. <pre>an3000-mcm-slot7cpu1(config)# services subscriber-firewall subscriber-firewall-flow <name> flow-state-tracking-profile-selection global</pre> 2 Configure a flow-timeout-filter. See the <i>Affirmed Networks Gateway Administration Guide</i>. 3 Assign the flow-timeout filter to the flow-idle-timeout-rule that defines the timeout values. <pre>an3000-mcm-slot7cpu1(config)# service-construct flow-idle-timeout-rule <name> flow-idle-timeout-filter <name></pre> 4 Configure the timeout values for the connection states in the flow-idle-timeout-rule. <pre>an3000-mcm-slot7cpu1(config)# service-construct flow-idle-timeout-rule <name> tcp-open-timeout <seconds> tcp-established-timeout <seconds> tcp-close-timeout <seconds> udp-open-timeout <seconds> udp-established-timeout <seconds></pre>

Feature	Description
Support for monitoring the different establishment states of TCP and UDP flows accepted by a subscriber firewall flow (continued)	5 Assign the flow-idle-timeout-rule to a flow-idle-timeout-profile. <pre>an3000-mcm-slot7cpu1(config)# flow-idle-timeout-profile <name> flow-idle-timeout-rule <name></pre> 6 Assign the flow-idle-timeout-profile to the data profile. <pre>an3000-mcm-slot7cpu1(config)# workflow data-profile <name> flow-idle-timeout-profile <name></pre> All subscriber firewall flows that enable global monitoring use this flow-idle-timeout-profile. To locally configure monitoring of the flow establishment states: 1 Enable local control for the tracking profile selection. <pre>an3000-mcm-slot7cpu1(config)# services subscriber-firewall subscriber-firewall-flow <name> flow-state-tracking-profile-selection local</pre> 2 Configure the timeout values for the connection states in a flow-state-tracking-profile. <pre>an3000-mcm-slot7cpu1(config)# flow-state-tracking-profile <name> tcp-open-timeout <seconds> tcp-established-timeout <seconds> tcp-close-timeout <seconds> udp-open-timeout <seconds> udp-established-timeout <seconds></pre> 3 Assign the flow-state-tracking-profile to the subscriber-firewall-flow. <pre>an3000-mcm-slot7cpu1(config)# services subscriber-firewall subscriber-firewall-flow <name> flow-state-tracking-profile <name></pre>

Feature	Description
Added support for clearing NAT statistics	This feature adds commands for clearing NAT statistics. The only statistics that will not be cleared are concurrent counters, which display NAT resources that are currently being used in the MCC. For example, concurrent counters include connection-count and resource-utilization-session. The following command clears NAT statistics for all the nodes: <pre>services subscriber-firewall nat cluster-level module-statistics <chassis-id> clear</pre> The following command clears NAT statistics for a particular node: <pre>services subscriber-firewall nat node-level module-statistics <chassis-id> <node-id> clear</pre> The following command clears NAT statistics for a statistical nat-pool on a particular node: <pre>services subscriber-firewall nat statistical nat-pool-statistics <node-id> <network-context> <nat-pool> clear</pre> The following command clears NAT statistics for a sub-pool of a statistical nat-pool on a particular node: <pre>services subscriber-firewall nat statistical sub-pool-statistics <node-id> <network-context> <nat-pool> <sub-pool> clear</pre> The following command clears NAT statistics for a deterministic nat-pool: <pre>services subscriber-firewall nat deterministic nat-pool-statistics <network-context> <nat-pool> clear</pre> The following command clears NAT statistics for a sub-pool of a deterministic nat-pool: <pre>services subscriber-firewall nat deterministic sub-pool-statistics <network-context> <nat-pool> <sub-pool> clear</pre> The following command clears NAT statistics for a guaranteed nat-pool: <pre>services subscriber-firewall nat guaranteed nat-pool-statistics <network-context> <nat-pool> clear</pre> The following command clears NAT statistics for a sub-pool of a guaranteed nat-pool: <pre>services subscriber-firewall nat guaranteed sub-pool-statistics <network-context> <nat-pool> <sub-pool> clear</pre>
Added support for a predefined service rule to be activated under a dynamic rating group and for a predefined service rule under a rating sub-group to be moved to a dynamic rating group based on PCRF messaging. See the <i>Affirmed Networks Gx Interface Specification</i>.	This feature allows for a predefined service rule to be activated under a dynamic rating group. For example, if the quota-id received in the Charging-Rule-Definition AVP is not pre-configured on the MCC, the MCC will dynamically allocate a rating group with the received quota-id. If the Charging-Rule-Definition AVP also contains the name of a predefined service rule, the MCC will activate the predefined service rule under the dynamic rating group. This feature also allows for a predefined service rule under a rating sub-group to be moved to a dynamic rating group based on PCRF messaging. Note: Ensure the dynamic rating group's quota ID is not configured on the MCC.

Feature	Description
Factor additional latency for distance for congestion detection	This feature adds a configuration option for Round-trip Time (RTT) offsets when RTT is used in the congestion detection's throughput formula. RTT offsets can be configured in data centers for both <i>nearby</i> and <i>far away</i> subscribers to avoid misidentifying <i>far away</i> subscribers as congested. This feature reports the length of time that the client connection remains in each congestion level in the HTDR/EDR and also reevaluates policies for flows where the congestion levels change. This feature adds the following new HTDR/EDR fields in tcpClientConnectionInfo: ■ CongestionLevelNoneTime ■ CongestionLevelMildTime ■ CongestionLevelModerateTime ■ CongestionLevelSevereTime To configure RTT offsets: <pre>an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy rtt-location-offset-profile RTT-PROFILE</pre> Possible completions: <pre>compromise-rtt-offset RTT location offset for subscribers without a known TAC location-offset RTT location offset for subscribers with a known TAC</pre> <pre>an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy rtt-location-offset-profile RTT-PROFILE location-offset RTT-LOCATION1</pre> an3000-mcm-slot7cpu1(config-location-offset-RTT-LOCATION1)# ? Possible completions: <pre>rtt-location-offset RTT location offset(ms) tac-high-byte high byte of TAC</pre> <pre>an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance USTCP tcp-measurement</pre> Possible completions: <pre>admin-state Enable/Disable TCP Measurement congestion-level Congestion Level for applying Optimizations based on Client Side TCP measurement rtt-location-offset-profile RTT Location Offset Profile for this instance sampling-interval Client Side TCP measurement Sampling Interval (1/ms)</pre>

Feature	Description
Added support for the VM name to the TFDR and HTDR Recording Entity field.	If the vm-name is configured for a cluster node, the vm-name is included in the Recording Entity field in TFDRs and HTDRs. If the vm-name is not configured for a cluster node, a generic string, such as node_personality-node_number, will be used for the vm-name in TFDRs and HTDRs. For example, ISM-5, CSM-1, ASM-2, etc. The HHE header has a hard-coded limit of 2048 bytes. If the header addition with the vm-name exceeds this limit, the actual value in the header will get truncated to fit in 2048 bytes.
Added the ISM VMNAME on which the session has landed	Use the instructions in the <i>Affirmed Networks System Administration Guide</i> to configure the vm-name. If the vm-name is not configured, MCC uses a generic name. For example: <pre>an3000-mcm-slot7cpu1(config)# cluster <id> node <id> vm-name</pre> This feature adds the ISM VMNAME in the header. For example: <pre>an3000-mcm-slot7cpu1(config)# service-construct http-header-enrichment name request add-header header-string name value</pre> Possible completions: <pre><String, may include variable such as IMSI, MSISDN, HTTPVERSION, UEIPADDR, ORIGINALAPN, IMEI, ULICGI, RATTYPE, GWIPADDR, PLMNID, CHARGINGCHAR, TIMESTAMP, SUBSCRIBERID, VMNAME, example: \$(MSISDN)></pre>
Added support for specifying how the IP type is chosen when PGW-originated messages are sent to the ePDG See the <i>Affirmed Networks Gateway Administration Guide</i>.	This release adds CLI parameters that let you specify how the IP type (IPv4 or IPv6) is chosen when the PGW originates messages that are sent to the ePDG. In previous releases, the PGW always used IPv6 when it originated messages that it sent to the ePDG. To configure how the IP type is chosen, use the following command: <pre>an3000-mcm-slot7cpu1(config)# zone default gateway pgw <gateway-name> gtpc-endpoint-params dual-stack-behavior [initial-ip-type-used-by-peer ipv4 ipv6]</pre> where: ■ initial-ip-type-used-by-peer – The PGW should use the IP type that was used in the Create Session Request from the ePDG. ■ ipv4 – The PGW should use IPv4. ■ ipv6 – The PGW should use IPv6.

Feature	Description
Added ability to select an online-charging interface using the CRBN profile and to configure the time-threshold and volume-threshold for the online-charging interface See the <i>Affirmed Networks Gy Interface Specification</i>.	This feature allows you to select an online-charging interface using the CRBN profile if that CRBN profile has an online-charging interface configured. <pre>an3000-mcm-slot7cpu1(config)# services crbn-profile <profile-name> online-interface <interface-name></pre> Note the following guidelines when using this feature: ■ The online-charging interface configured for the CRBN profile overrides the online-charging interface configured for the workflow control profile. ■ If the MCC gateway receives multiple CRBN profiles from the PCRF or RADIUS server, the MCC uses the first CRBN profile in the list to select the online-charging interface. This feature also adds two new online-charging configuration options to configure the time-threshold and volume-threshold. The MCC uses these configured values when the OCS does not send the Time-Quota-Threshold and Volume-Quota-Threshold AVPs within the MSCC AVP and these values are not configured under the rating group. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> attributes time-threshold <value> Volume-threshold <value></pre> Note the following guidelines when using this feature: ■ The time-threshold and volume-threshold configured for the rating group take precedence over the time-threshold and volume-threshold configured for the online-charging interface. ■ If the GSU is received with result code 4012, the MCC does not apply the time-threshold and volume-threshold configured for the rating group or online-charging interface.
Increased maximum MTU and MRU to 9100 bytes on all IP interfaces See the <i>Affirmed Networks Gateway Administration Guide</i> and the <i>Affirmed Networks System Administration Guide</i>.	IP interfaces on the MCC can now support an MTU and MRU as large as 9100 bytes. Prior to this release, the maximum MTU/MRU value was 1800. To configure the MTU/MRU size, first set the maximum packet size of the cluster with the new max-packet-size command: <pre>an3000-mcm-slot7cpu1(config)# cluster <id> max-packet-size <bytes></pre> This value has a default of 2048, and a range of 2048 through 9216. This value must be at least as large as the MTU value you want to use for the IP interfaces, and you must reboot the cluster. Next, set the MTU for IP interfaces: <pre>an3000-mcm-slot7cpu1(config)# network-context <name> ip-interface mtu <bytes></pre> This value has a default of 1500, and a range of 576 through 9100.

Feature	Description
Statistics support for non-200 HTTP responses	Added non-200 HTTP response statistics to the show services multi-protocol-proxy statistics [cluster-level node-level] detail command: Client Response Status Information: Client HTTP Response Status Information (1xx) Client Response Status Success OK: Client HTTP Response Status OK (200) Client Response Status Success: Client HTTP Response Status Other (2xx) Client Response Status Redirection: Client HTTP Response Status Redirection (3xx) Client Response Status Client Error: Client HTTP Response Status Client Error (4xx) Client Response Status Server Error: Client HTTP Response Status Server Error (5xx) Acuitas support for these statistics in the following statistics tables: CONTENTHTTPPROXYDETAILSTATSCHASSIS CONTENTHTTPPROXYDETAILSTATSSLOT
Added statistics on NAT connection failures	Added performance statistic at the APN level that indicates the total number of connection failures due to max-port-chunks exceeded for all sessions associated with that APN. show zone default gateway statistics cluster-level call-performance nat-conn-failure-max-port-chunks-exceeded Acuitas support for this statistic in the following statistics tables: APNCALLPERFSTATSGRID APNCALLPERFSTATSCHASSIS APNCALLPERFSTATSSLOT Added performance statistic at the NAT Pool level to track the cumulative number of port chunk allocation failures due to maximum chunks allocation limit for all sessions using that NAT pool. show services subscriber-firewall nat performance-statistics nat-pool nat-conn-failure-max-port-chunks-exceeded Acuitas support for this statistics in the following statistics table: NATPOOLPERFSTATSSLOT Added performance statistic at the NAT Pool level to track the per-session high watermark for the number of port chunks that were allocated on a specific slot. show services subscriber-firewall nat statistical nat-pool-statistics per-session-port-chunks-high-watermark

Feature	Description
ePDG support for sending HTTP POST request messages using a configurable URI in the HTTP server configuration	This feature adds support for a configurable URI field (uri-path) in the HTTP server configuration. The ePDG uses the HTTP protocol to communicate subscriber information (IMSI) to a provisioning server. When the ePDG receives a Diameter-EAP-Answer (DEA) response from the AAA server with the result code, 5450-DIAMETER_ERROR_USER_NO_NON_3GPP_SUBSCRIPTION, the ePDG sends an HTTP_POST request with the subscriber IMSI included in the URL to the provisioning server. Configure the URI path: <pre>an3000-mcm-slot7cpu1(config)# service-construct http-server <name> uri-path <string></pre> Example: <pre>an3000-mcm-slot7cpu1(config-http-server-HTTP_TEST)# uri-path /soa-infra/services/Custom2D/ CreateVOWIFProvisioningEPDGReqABCSImpl/HTTPPOSTCreateVOW IFIProvisioningIMSI=\$(IMSI)&operationName= Request-Response</pre> where: ■ uri-path is the server URI path. The server-ip-address and server-port values are added to the uri-path to create the URL for HTTP POST. If the uri-path is not configured, the request message uses the default URI format /imsi/\$(IMSI). ■ <string> values include: IMSI, MSISDN, HTTPVERSION, UEIPADDR, ORIGINALAPN, IMEI, ULICGI, RATTYPE, GWIPADDR, PLMNID, CHARGINGCHAR, SUBSCRIBERID, for example: \$(MSISDN). Associate the http-server-list with the appropriate http-server: <pre>an3000-mcm-slot7cpu1(config)# service-construct http-server-list <name> http-server <name></pre> Associate the http-server-list with the ePDG: <pre>an3000-mcm-slot7cpu1(config)# zone default gateway epdg <name> provisioning-server-list <server-list-name></pre>

Unsupported Features

This section lists the unqualified or unsupported features in this release. The following features may appear in the documentation and in Acuitas and the CLI, but are not fully supported in this release:

- OSU from releases prior to 9.2.2.2.
Note: OSU from 9.2.2.2/9.3.0.2 and later to 9.4 is supported.
- ePDG and TWAG is not supported in the 9.3 and 9.4 releases.
- In clusters with ISMs, user-space-networking is always enabled. Kernel-based networking is not supported.
- Geo-redundancy with pre-9.2 releases is not supported. Upgrade of pre-9.2 geo-redundant clusters will lose sessions.
- content-cache with ISM and user-space-networking.
- Data profile switching (for example, triggered by MVNO, CC group, RAT change, etc.) along with CRBN profile functionality.
- CRN and CRBN cannot activate the same service rule.
- Scale In/Scale Out is not supported for CUPS and non-CUPS for all VM personalities.
Note: Manual Scale Out is supported only for ASMs/CPMs/SSMs in a CUPS cluster.
 - The **disabled** and **maintenance** options for **cluster <id> node <id> admin-state** are not supported. See [Update to Configuration Options for cluster <id> node <id> admin-state \(page 85\)](#).
- In the **protocol-application-rule-group**, only the flow attribute values of audio and video are qualified.
- Support for the **apn-reporting** parameter under the RADIUS authorization and accounting interfaces has not been qualified.
- 5G NSA: Modify Access Bearer Request messages.
- ISM back-end forwarding (BEF).
- SR-IOV base devices when management devices are non-SR-IOV is not qualified for RedHat OpenStack or VMware.
Note: SR-IOV base devices when management devices are non-SR-IOV is supported for Ubuntu OpenStack Newton.
- VoLTE SIP statistics are not supported on standalone SGW systems:
`an3000-mcm-slot7cpu1# show data-path statistics sip-statistics`

- MCC does not support geo-redundancy for GiGW/WAG sessions and GTPv0 sessions.

Unsupported CUPS Features

- IPv6 Prefix Delegation.
- UPF geo-redundancy.
Note: CPF geo-redundancy is supported.
- GTP-U F-TEID allocation on CPF.
- Rf-Gy sync is not supported for CUPS.
- 2G GTP v0 calls are not supported on CUPS; specifically, GTP v0 sessions on the UPF are not supported.
Note: 2G GTP v1 calls are supported on CUPS/UPF. Also, 2G GTP v0 calls can be terminated on the CPF as non-CUPS calls if GTP-U interface(s) are available on the CPF.
- UPF selection based on DNS query.
- UPF selection based on dynamic UPF load.
- Dual stack: IP v4/v6 is not supported on the PFCP (Sx) interface.
- Standalone Sxc, that is, standalone TDF-C and TDF-U.
- Co-located SGW-U and PGW-U sessions when selected by different/independent SGW-C and PGW-C, that is two separate PFCP establishments for the same subscriber on the same UPF.
Note: Co-located SGW-U + PGW-U when established by the same CPF (single PFCP Session Establishment) is supported.
- Third-party CPF or UPF.
- Limited support for PFCP IEs.
- PFCP PFD Management Procedure.
- PFCP Node Report Procedure.
- PFCP Overload Control Procedure.
- Co-located CPF and UPF in the same MCC cluster.
- Insertion strategy from SAEGW to SAEGW + SAEGW-U.
- GGSN service is not supported for CUPS.

Note: PGW with GGSN support enabled is supported for CUPS.

- Flow authorization interface is not supported for CUPS.

Unsupported TAM Features

- OSPF (Open Shortest Path First).
- TAM is not supported on the following products: GiGW, ePDG, TWAG, GTP Proxy.
- CUPS and non-CUPS UPF clusters do not support a mix of TAM and non-TAM sessions.
- Geo-redundancy.

Unsupported CALEA Features

- *CALEA Multiple X2/X3 Delivery Destinations* with non-CUPS sessions do not delete dynamically created X3 interface-groups and interfaces which are no longer in use by any intercepted session.
- Packet Data Header Information Reporting (PDHIR) on CUPS systems configured for *Call Content from CPF LI Targets*.

Note: The PDHIR feature is supported only on CUPS systems configured for *Call Content from UPF Static LI Targets* or *Call Content from UPF Dynamic LI Targets*.

User Account Access to the Network Element

This section describes the user accounts/passwords supported in this release.

User Account Access	Description
■ root/root ■ admin/admin ■ emsadmin/emsadmin ■ calea/calea ■ content/content, user3g/user3g, user4g/user4g	Provides the following access to the chassis: ■ ssh access on port 22 to the CLI on the chassis ■ scp/sftp access on port 2222 to the chassis The admin account can also create other sftp/CLI accounts (other than group calea accounts) using CLI commands. The calea account can also create other calea group sftp/CLI accounts using CLI commands. Note: After a new installation, the network element must be initialized using the <i>installation and initialization</i> script. Execute the script (<i>/opt/thoroughbred/an3000-init</i>) through the console port on an MCM. Affirmed Networks recommends enabling password aging by answering Y to the following prompt: <pre>Enable password aging? <Y/n></pre> When you enable password aging, you are prompted to change the password on the first login, and again 45 days later. Affirmed Networks recommends disabling the user root by answering N to the following prompt: <pre>Enable root user ssh access (strongly recommend no)? <y/N></pre> To enable ssh, see Getting Started (page 50).
maint/maint	Provides maint access on port 2222 to the chassis. The maint group account is a non-root account used to log into base Linux system(s). The maint group account is intended for use by Operators and Affirmed Networks Engineers for basic system maintenance and troubleshooting on systems where root access is not enabled (typically production systems). Users assigned to a maint group account can observe and maintain the MCC without compromising its integrity or security. For example, maint group users can perform maintenance-level commands such as reboot, rollback, and tos start/stop as well as perform analysis and view logs. See About the maint Group Account (page 49) for details. Note: Affirmed Networks recommends normally disabling maint group ssh access in field deployments. Enable maint group ssh access only for brief time periods in which maint access is desired and expected.
admin-maint	As a hybrid user, remote TACACS+ users can log in to both the CLI and the Linux shell using the group admin-maint in the TACACS+ custom AV pair in the initial authorization.

About the maint Group Account

The **maint** group account is a non-root account used to log into base Linux system(s). The **maint** group account is intended for use by Operators and Affirmed Networks Engineers for basic system maintenance and troubleshooting on systems where root access is not enabled (typically production systems). Users assigned to a **maint** group account can observe and maintain the MCC without compromising its integrity or security. For example, **maint** group users can perform maintenance-level commands such as reboot, rollback, and tos start/stop as well as perform analysis and view logs.

Commands issued by **maint** group users are logged to the messages file.

Note: *If additional user privileges or root-like commands are required, contact the TAC.*

About maint Users

There are two types of **maint** users:

*default **maint*** – The default **maint** account is created during system startup and is resident on the system. The default password is **maint** and the login is **maint**. The default **maint** user has additional privileges that other **maint** group users do not. See [Managing Default maint Users \(page 52\)](#).

*additional **maint** users* – The additional users (**maint** group users) created by the default **maint** user.

Getting Started

By default, ssh access is disabled. To enable ssh access and establish an ssh session, log in to the MCM console and issue the **sshenable** command as follows:

***Note:** Affirmed Networks recommends disabling ssh access in field deployments.*

- 1 Log in to the active MCM console as **maint** and issue the **sshenable** command to allow ssh access to all maint group users (if desired).

***Note:** The **sshenable** command must be issued on the active MCM.*

- 2 (Optional) Issue the **maintadmin** command to add additional **maint** group users.
- 3 Issue the **sshdisable** command from any **maint** group user account to temporarily disable ssh access until further ssh access is desired. Console access will still be available.

Managing maint Group Users

All **maint** group users can access non-root Linux bash commands to view logs, releases, and configuration information; however, **maint** group users do not have modify permissions. **Maint** group users can issue the following Linux commands:

Command	Description
maintgroup	Lists the names of the maint group accounts.
w(who)	Search for logged-in maint group users.

Maint group users can also issue the following commands as root to allow certain access to the system:

Command	Description
tos	Arguments include: start, stop, hist, status, and restart.
sshenable	Enables ssh connections for all maint group users. See Getting Started (page 50) . <i>Note: The sshenable command must be issued on the active MCM. When you enable ssh access for maint group users, that access applies to the entire cluster across reboots and upgrades.</i>
sshdisable	Disables ssh access for all maint group users and terminates all ssh sessions owned by all maint group users. This command does not terminate console sessions by maint group users. See Getting Started (page 50) . <i>Note: You must issue the sshenable command on the active MCM.</i>
reboot	Reboots the specific card. <i>Note: Issuing the reboot command on the active MCM results in an MCM switchover.</i>
rollback.py	Issue this command on both MCMs. If this command is issued after a successful tostop, the current release running on this card reverts to the prior release (if one exists). <i>Note: Use this procedure only if the image-management-rollback CLI command is unavailable.</i>
chassis_reboot	Issue this command on the active MCM to reboot the system.

Managing Default maint Users

The default **maint** user is the only user that can issue the **maintadmin** command. This command enables the **maint** user to create and delete other **maint** group users (but not itself) and set **maint** group passwords.

Note: You must issue the **maintadmin** command on the active MCM to automatically replicate changes to all cards in the system.

The **maintadmin** command supports the following options:

Option	Description
-create	create <user> in maint group
-remove	remove <user>
-set	set <user> attributes including password
-user <user>	specify user name
-password <password>	specify password
-passwordorpk	allow password <u>or</u> pki ssh authentication
-pkionly	allow only pki ssh authentication

Note: If you did not configure the password when creating a user, the initial default password is the username. For example, for username **user1**, the default password is **user1**.

Network Considerations

Operators should take into account the following network considerations prior to installing, configuring, or upgrading Affirmed Networks products.

Issue	Description
All the fragments of a packet must be received on the same IOM	When a packet fragment arrives on a vIOM or on the IOM function of a vSSM or vISM, it is queued. All the fragments for the packet must be received on the same IOM to successfully reassemble the packet and forward it for additional processing. If the fragments are received on multiple IOMs, reassembly fails and the fragments are dropped. The MCC peer router must be configured to route all of the packet fragments identically.
Do not use node number 1 for SSM deployments	Proxy traffic is dropped for some sessions by the Linux Kernel on the CSM if the SSM uses the network address 169.254.128.0 as its internal IP address. Workaround: <i>Redeploy the SSM and do not use node number 1 for SSM deployments.</i>
Modifying NTP servers	Modifying (adding or deleting) the list of configured NTP servers using the CLI causes the NTP daemon to restart which may result in a brief assertion of the <i>anNtpSyncError</i> alarm. The alarm automatically clears when the system is in sync. Note: If you delete the last configured NTP server, the system asserts the <i>anNtpSyncError</i> alarm and this alarm does not clear.
Cluster ID greater than 127 is not supported	MCC supports only cluster IDs up to 127. Trying to configure a cluster ID greater than 127 will result in the [ERROR]: this is an invalid value error message.
Set the IP OSPF interface to 1500 for OSPF over GRE	In order to establish an OSPF adjacency, both OSPF routers must advertise the same MTU in their Database Description packets. When OSPF runs over a GRE Tunnel, OSPF does not advertise the MTU of the physical interface that anchors the Tunnel. The advertised MTU is adjusted by the tunnel overhead (size of IP + GRE headers). If the peer OSPF router does not make the same adjustment to its advertised MTU, adjacency is not established. Workaround: <i>Perform either of the following workarounds:</i> ■ Under ip interface ip-ospf settings, enable mtu-ignore. Note: <i>This procedure must also be performed on the peer router.</i> ■ Under ip interface level ip-ospf settings, set the OSPF MTU to match the peer router.
VM BIOS: Disable the power management setting	If the virtual machine BIOS setting has a power management setting, make sure it is turned off. See the appropriate deployment guides for HP systems. See How to Obtain Product Documentation (page 4) for details.

Issue	Description
Deep Packet Inspection CLI considerations	If you make any changes to the list of available DPI protocols, you must issue a soft boot of the SSM30 card for the change to take effect. <pre>chassis <chassis_number> card <ssm_slot_number> reboot soft</pre> <i>Note: The list of protocols is used to initialize the DPI detection engine when the fastpath application on the SSM30 is started or restarted. By restricting the set of protocols, the CPU cycles expended for DPI detection are reduced. De-selecting protocols selected by one or more application-rule-group objects are rejected</i>
Modification of online interface (OCS)	Modifying online interface attributes during an active session may cause some of the attributes to take effect on that session. Instead, the existing session should continue with the attributes that were present prior to modifying the online interface attribute. <i>Workaround: Modify the online interface only after verifying that no session is using that online interface. Modifying the online interface during an active session might cause the system to send different attributes than expected.</i>

Restrictions and Special Considerations

The following restrictions and special considerations apply to this release.

Restriction/Special Consideration	Description
Do not rollback/downgrade to a DPI signature prior to dpisig_2019-10-18	Operators must not rollback/downgrade to a DPI signature prior to dpisig_2019-10-18 because previous DPI signature versions are not compatible with MCC release 9.4. Performing a rollback/downgrade will cause an SSM fault because it requires CDC 2.0 and the NAT v2 plugin.
Signaling statistics for CUPS	All signaling statistics that pertain to GTP-C are available only on the CPF. On the UPF, these statistics are displayed as zero.
Data-usage statistics for CUPS sessions	For CUPS sessions, data-usage statistics, such as call-performance uplink/downlink packet/byte counts, deep-packet-inspection statistics, and any other data profile services statistics (for example, proxy, content-filtering, and other vas-service statistics) are available only on the UPF. On the CPF, these statistics are displayed as zero.
Do not use hypervisor-level commands to pause/unpause VMs	Do not use hypervisor-level commands to pause/unpause or suspend/resume virtual machines (VMs). Affirmed Networks does not support these commands because they may cause unintended events such as a cluster fault/reboot.
Rate limiting support	Rate limiting of up to 25% only of total proxy throughput is supported.
Network context scaling for RADIUS peers configured with Virtual Routing and Forwarding (VRF) lightweight network contexts.	Network context scaling for RADIUS peers configured with Virtual Routing and Forwarding (VRF) lightweight network contexts is CPU intensive and involves a significant number of message transactions that may impact GwServiceDirector, NetDirector, NetAgent, NAS, PDNGW, SGW, RADIUS, and FastPath tasks. Affirmed Networks recommends scaling the configuration in a controlled manner or diverting traffic only after all of the new configurations have successfully completed.
Modifying update-source in shared subnets for BGP peering	For shared subnets, modifying update-source after a BGP peer is created is not supported. Operators must first delete the BGP peer, modify update-source, then reconfigure the BGP peer for changes to take effect.
Reduction of E/W traffic using TAM or ISM/IWM VMs	In certain cases, reduction of E/W traffic using TAM or ISM/IWM VMs is precluded. The E/W traffic reduction does not occur in the following scenarios: ■ Proxy services result in E/W traffic if SSMs are used. ■ E/W traffic in TAM mode is not reduced when a VAS is used if SSMs are in the cluster. ■ IPsec tunnels, GRE VPNs, or GRE/L2TP tunnels may result in East/West traffic.

Restriction/Special Consideration	Description
CUPS session data-usage statistics	For CUPS sessions, data-usage statistics, such as call-performance uplink/downlink packet/byte counts, deep-packet-inspection statistics, and any other data profile services statistics (for example, proxy, content-filtering, and other vas-service statistics) are available only on the UPF, not the CPF.
Maximum of 30 SDCs can be included in Gz/Rf CDRs	A maximum of 30 service data containers (SDCs) can be included in Gz/Rf CDRs and any additional SDCs are dropped.
TWAG support for Context-Identifier AVP in Diameter- EAP-Answer (DEA) on STa interface	With Release 9.2, the TWAG/TWAP, by default, processes the Context-Identifier AVP when provided by the AAA server in the Diameter- EAP-Answer (DEA) message to select an APN for the session. The Context-Identifier AVP cannot be suppressed when received in the DEA message.
Upgrade consideration for modified rating group memory allocation when a billing plan is applied to a subscriber	Rating group memory allocation is different in 9.2, which may result in some service rules not being loaded during OSU from 9.1 to 9.2. Therefore, OSU from 9.1 to 9.2 is not supported.
An existing application-rule-group in MCC releases prior to 8.4 <i>cannot</i> be automatically converted to the MCC release 8.4 protocol-application-rule-group configuration	Due to architectural changes for the new DPI classification model in MCC release 8.4, an existing application-rule-group in MCC releases prior to 8.4 <i>cannot</i> be automatically converted to the MCC release 8.4 protocol-application-rule-group configuration. Important steps are required before and after upgrading from a release prior to 8.4 to the 8.4 or later release. - Before the upgrade, issue the following command to save the existing configuration: <pre>an3000-mcm-slot7cpu1# show running-config save test.cfg</pre> The configuration file is saved to the path <pre>./boot/partitions/current/home/users/admin/test.cfg</pre> - Perform the upgrade to MCC release 8.4. Be aware that MCC will experience loss of DPI functionality after the upgrade until you configure the new protocol-application-rule-group. - Following the upgrade, configure the new protocol-application-rule-group using the data in the test.cfg.
Single ePDG Geo-Group Per SWu Network Context	When configuring a geographic-redundancy group (geo-redundant-group) on the ePDG, it is important to note that only a single geo-redundant-group can be configured in a dedicated SWu network context with security enabled on the IPsec tunnel. To configure a geo-redundant-group: <pre>an3000-mcm-slot7cpu1(config)# zone <name> gateway epdg <epdg-name> fully-qualified-domain-name <FQDN> geo-redundant-group</pre>

Restriction/Special Consideration	Description
Clearing DPI statistics	Clearing any of the DPI statistics returns a confirmation prompt indicating that the action clears all of the DPI statistics. For example: <pre>zone <name> deep-packet-inspection statistics grid-level general clear</pre> This will clear all of the DPI statistics. Do you wish to proceed? (yes/no) : Any scripts that clear DPI statistics must account for this new prompt.
23091 Supported IXGBE driver versions	The following IXGBE drivers are supported: ■ IXGBE version 4.4.0-k-rh7.3 is supported for MCC RHEL 7.3 installations. ■ IXGBE version 3.21.4.3iov is supported for MCC VMware ESXi installations.

Restriction/Special Consideration	Description
28475 Using WindRiver dpdk pktgen 2.8.0	WindRiver PktGen is included in MCC software images. The following is the WindRiver license. Affirmed Networks distributes the binary version (2). Windriver license requirement <i>Copyright (c) 2010-2014, Wind River Systems, Inc.</i> <i>Redistribution and use in source and binary forms, with or without modification, are permitted provided that the following conditions are met:</i> <i>1 Redistributions of source code must retain the above copyright notice, this list of conditions and the following disclaimer.</i> <i>2 Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution.</i> <i>3 Neither the name of Wind River Systems nor the names of its contributors may be used to endorse or promote products derived from this software without specific prior written permission.</i> <i>4 The screens displayed by the application must contain the copyright notice as defined above and can not be removed without specific prior written permission.</i> <i>THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT HOLDER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.</i>
Efficiently using the address space defined in your NAT pools	To efficiently use the address space defined in your NAT pools, Affirmed Networks recommends setting the max-nat-workflow-modules parameter to coincide with the number of SSMs/WSMs in your cluster. In other words, in a cluster with only three SSMs/WSMs, set this parameter to 3 even though this parameter defaults to 15.
37767 Online Software Upgrade (OSU)	OSU (for GiGw/TWAG/ePDG) from 7.2.x -> 8.1/8.2 is not supported due to architectural changes related to 7.3.x georedundancy features.
Georedundancy is not supported when upgrading to Release 9.4.0.0.	Due to architectural and feature changes in this release, georedundancy is not supported when upgrading to Release 9.4.0.0. Both the primary and secondary MCC clusters must be upgraded to Release 9.4.0.0 to support georedundancy.

Restriction/Special Consideration	Description
Implemented a change to the dpi-pre-detection setting Also see Change to dpi-pre-detection Setting (page 65).	In this release, the workflow data-profile dpi-pre-detection-max-bytes and dpi-pre-detection-max-pkts default setting is changed to 0 (disabled). If either of these objects were set to a non-zero value (enabled) in a prior release and you do not intend to use this functionality, you must manually set the value to 0 (disabled) prior to upgrading, otherwise the system retains the non-zero value after the upgrade. ■ dpi-pre-detection-max-bytes – Allow max number of bytes for DPI pre protocol and application detection ■ dpi-pre-detection-max-pkts – allow max number of packets for DPI pre protocol and application detection
Disk space	Due to changes in the size and method in which software is stored on the MCC, Operators may notice a change in available MCM memory for customer-generated cores, fees, and statistics. Automated file rotation protection is still in place and this generated information is handled in the same manner as prior releases.
VoLTE SIP statistics are not supported on standalone SGW systems	VoLTE SIP statistics are not supported on standalone SGW systems: an3000-mcm-slot7cpu1# show data-path statistics sip-statistics
26311 MCC may drop user data packet in this instance	If a UE transmits a user data packet before the session transitions fully from an idle to an active state, the MCC may drop the packet. The modify bearer request and response must be completed in order for the user packet to transmit successfully.
vCenter datastore name must be unique in vCenter	When you configure a datastore to create a VM, the datastore name must be unique. It cannot contain blank spaces or parentheses.
CALEA restriction on platforms running v-switch	CALEA may not work for all combinations of packet sizes. If this configuration is desired, contact the TAC.
25925 Deploying VMs with node ID 128	In order to expand a cluster from 32 VMs to 128 VMs, you must install the latest OVAs. The most recent OVAs allow a node ID of 128. Workaround: <i>Deploy the node manually, without using the EMS Commissioning tools or the VNF Manager.</i>
Geographic redundancy	Some geographic redundancy feature limitations exist in older releases when the MCC is paired with a different major release. While running in this mode, Affirmed Networks recommends not making any configuration changes to the paired MCCs. The limitation no longer exists after all MCCs are upgraded.
24121 MCC supports default zone only	MCC only supports a single zone for the cluster, called the default zone. Do not configure zones other than the default.

Restriction/Special Consideration	Description
23282 Using WindRiver Accelerated VSwitch (used in WindRiver Titanium and HP Helion Carrier Grade)	If MCC is installed on a windriver host and AVS (Accelerated vSwitch) with AVP port type, the correct driver is instantiated. The following is the Windriver license. Affirmed Networks distributes the binary driver (2). Windriver license requirement <i>Copyright (c) 2013-2015, Wind River Systems, Inc.</i> <i>Redistribution and use in source and binary forms, with or without modification, are permitted provided that the following conditions are met:</i> 1 <i>Redistributions of source code must retain the above copyright notice, this list of conditions and the following disclaimer.</i> 2 <i>Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution.</i> 3 <i>Neither the name of Wind River Systems nor the names of its contributors may be used to endorse or promote products derived from this software without specific prior written permission.</i> <i>THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT HOLDER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.</i>
17878 TACACS+: Not supported on IPv6 networks	The current MCC TACACS+ implementation supports IPv4 networks only; IPv6 is not yet supported. Workaround: <i>Configure the MCC TACACS+ service with IPv4 addresses.</i>
Online usage not synched between active/standby chassis	The OCS/PCRF subscriber quota usage is not periodically synchronized between two geographic redundancy sites. Due to this issue, if geo-switchover occurs, the Operator loses the quota usage counts from the last CCR-U/CCR-I report to o avoid continuous message flood between two geo -sites for this rate geo-site failure event. Workaround: <i>To limit the loss of data usage statistics, configure a smaller quota threshold value through PCRF/OCS/Local config.</i>

Restriction/Special Consideration	Description
Workflow data-profile switching restrictions	The following workflow data profile switching restrictions apply to the qos-policy. ■ Data profile switching does not operate properly in the case of network initiated/user initiated dedicated bearers (i.e. configured locally to create dedicated bearers from the MCC). Data profile switching operates properly in the case of Dynamic bearers created by PCRF. ■ Data profile switching cannot convert a prepaid user to a postpaid user. The system is unable to initiate a new diameter session due to switching workflows.
Hot standby geo-redundancy	Currently, geo-redundancy is qualified for SGW/PGW/GGSN service with BGP IP backbone and one geo-redundancy group only. Support for geo-redundancy with OSPF backbone is not supported in this release.
Graceful restart in Gi NC results in BGP session reset in Gn NC	Graceful restart on any bgp or bgp-vpn router instance results in a graceful restart of all bgp and bgp-vpn instances running on the same CSM, not just the instance specified via the graceful restart command. Workaround: No workaround currently exists for this issue.
Change to the way in which IP subpools and IP loopback addresses are advertised	Due to the implementation of the georedundancy feature, Affirmed Networks no longer distributes UE pools and IP loopbacks as connected routes. Therefore, the ospfv2 router redistribute re-connected and ospfv3 router redistribute connected commands no longer advertise loopback and UE pool IP addresses. A change was made to modify the way in which IP subpools and IP loopback addresses are advertised by dynamic routing protocols such as OSPF and BGP. You are now required to configure redistribute static as part of your BGP, OSPFv2, and OSPFv3 configuration. For example: <pre>ip-protocols ospfv2 router GI router-id 11.24.2.2 redistribute static; top ip-protocols ospfv2 router GI interface gi-61 area 0.0.0.48; top commit</pre>
Adding default offline-interface to control-profile	The CLI requires a special format when adding a default offline-interface to a control-profile. When two or more entries exist, the entries must be enclosed in square brackets as follows: <pre>workflow control-profile controlprof default-offline-interface [CHARGINGLOCAL OFCSINT]</pre> This example adds the default-offline-interface CHARGINGLOCAL and OFCSINT.

Restriction/Special Consideration	Description
Using Internet Explorer to view notification messages	If you are using Internet Explorer to view the notification message configured for content filtering, do the following: 1 Open Internet Explorer. Click the Start and click Internet Explorer. 2 From the Tools icon, click Internet Options. 3 Click the Advanced tab and scroll down to the Browsing section. 4 Uncheck the Show friendly HTTP error messages check box and click OK.
BGP policies	BGP policies are used to control the routing information that is learned from a BGP peer and advertised to a BGP peer. If a BGP policy is modified, the peering sessions with the BGP peers to which this policy pertains must be restarted. For example: <pre>ip-protocols bgp[-vpn] router <network-context> clear all hard soft (in out both)</pre>
Requirement to change password on Acuitas server	If you change the password on the network element, you must also change that account information on the Acuitas server that manages that network element. To change the login password for the NE on the Acuitas server: 1 Right-click the network element and click Edit Element. 2 In the CLI & NETCONF section, enter the username and new password. 3 In the SFTP section, enter the username and new password. 4 Click Save. 5 At the confirmation prompt, click OK. See the <i>Acuitas User's Guide</i> for details.
In the case of a network initialized Dedicated Bearer, the HTTP UL traffic is sent on the dedicated bearer but the HTTP DL is sent on the default bearer	When the first packet in a flow for a given session is received, the packet will match on the appropriate bearer for that session. If a new, more appropriate bearer is signaled after this classification, the packets in the flow continue to use the original bearer. Flows started after the bearer is signaled would then match on the correct bearer. Workaround: <i>No workaround exists for this issue. It is unlikely to occur as bearers are generally signaled before data starts to flow. If this issue does occur, packets in the flow continue to use the original bearer until that flow becomes idle and ages out, therefore causing a new classification to occur.</i>

Restrictions for GQUIC User Agent Fingerprinting

- When an iOS device is connected to a primary device (Android) over Wi-Fi, it connects using proxy settings shared by the PdaNet application. As a result, all QUIC-generating applications on the iOS device revert to TCP. Therefore, GQUIC user agent fingerprinting is unable to detect tethered iOS devices.
- When an iOS device is connected to a primary device (Android) over Bluetooth, GQUIC traffic generated from the iOS device does not contain a UAID. Therefore, GQUIC user agent fingerprinting is unable to detect tethered iOS devices.
- Flow-level tethering detection is not supported for GQUIC user agent fingerprinting. Only the subscriber-level tethering status is detected by GQUIC.
- Tethering is not marked as true when the same UAID is present in GQUIC traffic generated from the same application on the primary and tethered devices. For example, if YouTube traffic from the primary device has GQUIC UAID as "Chrome/68.0.3440.91 Android 7.0; SM-G615F Build/NRD90M" and the same UAID is detected from the tethered device's YouTube traffic, GQUIC user agent fingerprinting cannot detect the tethering status.
- Tethering detection using GQUIC user agent fingerprinting requires GQUIC traffic to be generated by both the primary and secondary devices. For example, applications such as YouTube, Snapchat, Play Store, Google Drive, Google Search, and Google Maps should be used on both the primary and tethered devices to generate GQUIC traffic.

Configuration to Control DPI Decoding for Persistent HTTP Connections

For HTTP persistent connections, a single TCP connection is used to send multiple HTTP requests/responses. Because the persistent connection is not closed, all packets are subjected to DPI decoding, which can cause an increase in CPU usage.

To avoid a CPU usage spike, configure the following new parameter:

data-path dpi decode-first-http-transaction-only [enabled | disabled]

- **enabled** (Default) – On persistent HTTP connections, as soon as fastpath detects the first server response header, the connection is closed (at L7 level) and subsequent HTTP transactions *are not* sent to the DPI engine for decoding. This setting can cause classification errors, as DPI examines only the first transaction.
- **disabled** – On persistent HTTP connections, packets of all HTTP transactions are sent to the DPI engine for decoding.

Rename corefiles to fault

Affirmed Networks renamed **corefiles** to **fault**. This name change impacts filenames, traps, alarms, eventlogs, syslogs, and scripts used to capture faults and generate summary files and alarms.

Note: Depending on the MCC release you are running, **corefiles** is renamed to **fault** and **memimage** is renamed to **fault**.

Below are examples of this renaming change:

Old Name	New Name
cluster <id> node <id> processes core-dump	cluster <id> node <id> processes fault
infrastructure oom coredump	infrastructure oom faults
infrastructure corefiles -filter generation	infrastructure faults -filter generation
infrastructure corefile generation	infrastructure faults generation
filecommand: category name for /public/ corefiles	filecommand: category name for /public/ faults
file names under /var/corefiles/* core *	file names under /var/corefiles/* fault *
	Note: Existing /var/corefiles/* core * files are not renamed, only new files use the new naming convention /var/corefiles/* fault *
Alarm: anNew Core FileDetected	Alarm: anNew Fault FileDetected
save-tech-support-info script change: -- nocores	save-tech-support-info script change: -- nofaults
show cluster <id> node <id> crash -detail < core -filename>	show cluster <id> node <id> fault -detail < fault -filename>
show cluster <id> node <id> crash -summary <seconds-since-epoch> < core -filename>	show cluster <id> node <id> fault -summary <seconds-since-epoch> < fault -filename>
show cluster <id> node <id> coredump -summary	show cluster <id> node <id> fault -summary
cluster <id> clear crashes	cluster <id> clear- faults
show cluster <id> node <id> corefile -detail	show cluster <id> node <id> fault -detail

Change to dpi-pre-detection Setting

See the information in [Restrictions and Special Considerations \(page 55\)](#) and then continue with this section.

Prior to Upgrading:

- If you are using the dpi-pre-detection functionality, set the value to a non-zero number prior to upgrading. The system retains that value after the upgrade. For example, set the value to 30 packets and 6000 bytes for a conservative configuration.
- If you are not using the dpi-pre-detection functionality, Affirmed Networks recommends setting the value to 0 (disabled).

After Upgrading:

- Settings that were previously unspecified or not present are automatically changed to 0 (disabled) by default.
- Settings that were previously specified with a non-zero value are retained.

The following table displays the behavior of these settings before and after upgrading to the current release.

Setting and Behavior Prior to Upgrade		Setting and Behavior After Upgrade	
Unspecified or not present (default)	Enabled by default and both pkts and bytes are unlimited.	Values are set to 0 by default.	Functionality is disabled.
Both values are set to 0	Functionality is disabled.	Values prior to the upgrade are preserved and both values remain set to 0.	Functionality is disabled.
Non-zero value	Enabled with pre-detection max pkts and bytes as specified in the configuration.	Values prior to the upgrade are preserved and both values remain set with non-zero values.	Enabled with pre-detection max pkts and bytes as specified in the configuration.

Behavior Change to Modify and Delete Persistent CALEA Targets

A change introduced in MCC release 9.3 affects modifying and deleting persistent CALEA targets. As part of the upgrade to the ConfD database (MCC release 9.3) specifying the cleartext name (LIID) to modify or delete a Persistent calea target is no longer supported.

The name (LIID) is stored in encrypted form in the ConfD database:

- When creating the persistent target using the following config command, you must specify the name (LIID) in cleartext:
calea target-table <cleartext-name> ...
- After the target has been created, to modify or delete the target using the config command, you must specify the encrypted form of the name by using the **\$7\$<encrypted-name>** format of the name (LIID).

However, to avoid having to know the encrypted name of the target, it is highly recommended to instead use the action commands below to modify or delete the target:

If you do not know the **<encrypted-name>**, you must delete the target using the following action command:

calea deleteTarget liid <cleartext-name>

To modify a target, first delete the target using the following action command:

calea deleteTarget liid <cleartext-name>

Then, re-create the target using the new configuration.

Notes:

- In MCC releases 9.2 and earlier, the action command **calea deleteTarget liid <name>** applied only to non-persistent targets.
- In MCC release 9.3 and later the action command **calea deleteTarget liid <name>** now applies to both non-persistent and persistent targets.

Known Anomalies

This section lists the known anomalies in this release.

Critical Anomalies in 9.4.0.0

The following critical anomalies were found in this release.

Anomaly	Description
47188 When a rule moves from one bearer to another due to a rule modification, previously created SDFs cause the rating-group to enter an “invalid” state.	When a rule moves from one bearer to another due to a rule modification, previously created SDFs cause the rating-group to enter an “invalid” state. Workaround: <i>No workaround exists for this issue.</i>
47135 Some security policy rules fail to be installed in Fastpath when more than the maximum number of eight IP subpools are configured	Some security policy rules fail to be installed in Fastpath when more than the maximum number of eight IP subpools are configured for certain UE pools. The policy manager restricts the number of sub-pools addresses to eight to control the amount of memory used. Workaround: <i>Configure the security policy rule source-packet-filter and destination-packet-filter with an ip-prefix-length that can handle a large number of source/destination IPs with a single traffic selector entry.</i>
47344 Redirected traffic may be billed.	Redirected traffic may be billed if the subgroup charging method is not considered when counting packets against an SDF. Workaround: <i>No workaround exists for this issue.</i>
47376 Changes to the configured timezone do not take full effect until each node of the cluster, or the entire cluster, is rebooted.	Changes to the configured timezone do not take full effect until each node of the cluster, or the entire cluster, is rebooted. Specifically, the CDR field recordOpeningTime will not reflect a timezone configuration change until after a reboot. Workaround: <i>After changing the configured timezone, reboot the cluster. Alternatively, sequentially reboot each node in the cluster, ensuring each node is fully operational before proceeding to the next node.</i>
47262 During OSU, CLI displays “man-in-the-middle attack” warning.	During OSU from 9.3.2.1 to 9.4.0.0, CLI displays “man-in-the-middle attack” warning. Workaround: <i>Issue the pkissh -c clear learned command, or CLI equivalent, as each domain master on the reporting node.</i>
47517 Bearer identifier is changed after a CPM failover.	Bearer identifier is changed after a CPM failover. Workaround: <i>No workaround exists for this issue.</i>
47616 If a rule is moved to a new rating group, MCC does not fetch online quota.	If a rule is moved to a new rating group, MCC does not fetch online quota. Workaround: <i>No workaround exists for this issue.</i>

Major Anomalies in 9.4.0.0

The following major anomalies were found in this release.

Anomaly	Description
47431 When BGP instance neighbors are configured without update-source, enabling shared-subnet in network-context does not bring up all BGP sessions.	When BGP instance neighbors are configured without update-source, enabling shared-subnet in the network-context does not bring up all BGP sessions. Workaround: <i>Before enabling shared-subnet in network-context, if any BGP instances are present for the network-context, ensure all neighbors are configured with update-source.</i>
47246 Configuring more than 64 CALEA instances causes the CLI “caleaInterfaceIndexPool is full - No positions available” error message.	MCC supports 256 CALEA interfaces, but when more than 64 instances are configured, the CLI displays the “caleaInterfaceIndexPool is full - No positions available” error message. Workaround: <i>Configure a maximum of 64 CALEA interfaces on a single MCC.</i>
47223 MCC may not correctly intercept targets when the IMEI begins with a zero.	MCC may not correctly intercept targets when the target identity used is an IMEI that begins with a zero. Workaround: <i>Use another identity for the target.</i>
46990 When a PCC rule fails to install, output for the show subscriber pdn-session CLI command on the U-Plane still displays the rule.	In certain cases, when a PCC rule fails to install, output for the show subscriber pdn-session CLI command on the U-Plane still displays the rule under charging-rule-names. Workaround: <i>No workaround exists for this issue.</i>
46918 When no data is entered, pressing Enter at a yes/no CLI prompt is processed as yes.	When no data is entered, pressing Enter at a yes/no CLI prompt is processed as a yes (continue) to the command. Workaround: <i>Explicitly enter yes or no at the CLI prompt.</i>
46817 Proxy traffic fails over MPLS.	Proxy traffic fails over MPLS. Workaround: <i>If proxy services are configured, configure the MSS used to establish TCP connections based on layer-3 and the MPLS used to transport the packets. Specifically, IPv6 requires 20 bytes more than IPv4, and MPLS requires an additional 4 bytes.</i>

Anomaly	Description
46738 Multicast and configuring more than 8 VLANs on a virtual function is not supported for Fortville.	Multicast and configuring more than 8 VLANs on a virtual function is not supported for Fortville. Workaround: <i>Install the Intel i40e 18.9.21.1 driver to enable trust mode which supports multicast (OSPF) and more than 8 VLANs.</i> To install the tool to enable trust mode: <pre>esxcli software vib install --no-sig-check -v <full path>/vib vib - INT-ESX-intel-esxcli-650.1.0.0.1-4598673.vib /etc/init.d/hostd restart esxcli intel network sriovnic vf set -t on -v 1 -n vmnic0</pre> Repeat for all VFs in an SR-IOV NIC. To make this configuration permanent: Edit the <code>/etc/rc.local.d/local.sh</code> file and insert all of the <code>esxcli intel network sriovnic vf set -t on -v 1 -n vmnic0</code> Repeat for all VFs in an SR-IOV NIC.
46679 CPU utilization increases by 6-7% when flow-idle-timeout-filter is configured with a maximum of 1024 flow-idle-timeout-rules.	CPU utilization increases by 6-7% when flow-idle-timeout-filter is configured with a maximum of 1024 flow-idle-timeout-rules and the rules are ordered such that the matching rule/filter is the last rule (that is, the 1024th rule). Workaround: <i>Configure flow-idle-timeout-rules with the minimum number of rules required to match traffic.</i>
46637 UDP QUIC server reject count statistic is no longer updated.	The UDP QUIC server reject count statistic is no longer updated with QUIC protocol version Q046. Workaround: <i>No workaround exists for this issue.</i>
46054 Occasionally, redundant IPsec security associations (SA) are observed after OSU in a multi-CSM cluster.	Occasionally, redundant IPsec security associations (SA) are observed after OSU in a multi-CSM cluster. Workaround: <i>Configure DPD to remove the redundant SA when DPD is detected.</i>
47454 If all active rules in a subgroup type rating group (online enabled) are moved to a dynamic rating group along with a data profile switch, FINAL for the former rating group is not sent immediately at the profile switch.	If all active rules in a subgroup type rating group (online enabled) are moved to a dynamic rating group along with a data profile switch, FINAL for the former rating group (source rating group) is not sent immediately at the profile switch even though it did not have any active rules after the profile switch. FINAL for this rating group is reported at the end of the session. Workaround: <i>No workaround exists for this issue.</i>
46647 GNU tail process may not exit when the Affirmed CLI ping and traceroute commands complete.	The Affirmed CLI ping and traceroute commands use the GNU tail command, and the tail process may not exit when the Affirmed CLI ping and traceroute commands complete. Workaround: <i>If sh/tail process pairs accumulate on the active MCM, perform an MCM switchover to clear them.</i>

Anomaly	Description
47385 Operators require access to IRI Report records from the LEMF.	Operators require access to IRI Report records from the LEMF to make it easier to diagnose proper behavior of the Packet Data Summary Report (PDSR) feature Workaround: <i>Obtain the IRI Report records from the LEMF and compare them to the show subscriber pdn-session statistics.</i>
47074 CLI traceroute command does not support the --sport option.	CLI traceroute command does not support the --sport option, which allows Operators to specify the source port when the protocol is TCP. The kernel randomly chooses a port, which may not have been assigned to the CPM, and causes the IOM to forward the TCP response to the wrong slot or drop the response. Workaround: <i>No workaround exists for this issue.</i>
47393 TCP fast-aging-time of 30 seconds is not long enough for proxy to prevent binding reuse/duplicate with a kernel timeout 120 seconds.	TCP fast-aging-time of 30 seconds is not long enough for proxy to prevent binding reuse/duplicate with a kernel timeout 120 seconds. Fastpath uses CLI configured proxy idle-time and fast-aging-time to age out proxy flows (connection-idle-timeout for Fastpath flow). Workaround: <i>EDR validation should be done only after Fastpath flow is cleared.</i>
47557 PGW cannot disregard the GBR value when PCRF sends non-zero GBR/MBR values for a dynamic service rule attached to a non-GBR QCI.	PGW will perform one of the following actions when the PCRF sends non-zero GBR/MBR values for a dynamic service rule attached to a non-GBR QCI: ■ By default, PGW applies both values to the SDF and sends a UBR towards the GTP side. ■ Per a new configuration introduced in the fix for issue 47109, PGW ignores both MBR and GBR values for the SDF and categorizes the packet as yellow. Currently, PGW cannot disregard the GBR value for a dynamic rule attached to a non-GBR QCI when MCC receives non-zero GBR/MBR values. Also, PGW categorizes the packet as yellow when GBR is zero and MBR is non-zero. Workaround: <i>To apply a non-zero MBR for an SDF (attached to non-GBR QCI), the PCRF must send a non-zero MBR and a zero GBR for a dynamic rule attached to a non-GBR QCI. By default, PGW applies the non-zero MBR to the SDF and does not send a UBR towards the GTP side (packet is categorized as yellow).</i>
47499 During OSU, new session setups fail when multiple SSMs are rebooted.	During OSU, new session setups fail when multiple SSMs are rebooted. Workaround: <i>Perform OSU during a low-traffic window.</i>
43746 After a RedHat OpenStack Newton compute is power-cycled, occasionally, VMs are in an error state and do not restart.	After a RedHat OpenStack Newton compute is power-cycled, occasionally, VMs are in an error state and do not restart. Workaround: <i>Users must manually reboot the VMs. Upgrade to OpenStack Queens which contains the fix for this issue.</i>

Minor Anomalies in 9.4.0.0

The following minor anomalies were found in this release.

Anomaly	Description
47686 If the root user is locked out, the unlock timer does not allow access after the configured lock duration expires.	If the root user is locked out due to excessive password failures, the unlock timer does not allow access after the configured lock duration expires. Workaround: Log in as the maint user and manually unlock the root account.

Existing Critical Anomalies

The following critical anomalies found in an earlier release also exist in this release.

Anomaly	Description
46367 Packets are being incorrectly matched and dropped.	Traffic should be allowed for it to match specific service rules that have HTTP rule groups and packet filters configured. For service rules to be matched, DPI needs to receive some packets to be able to return its identified results. Since the default rule is to drop packets if they are not matched, DPI is unable to return any packets. Workaround: Set <i>dpi-pre-detection-max-bytes</i> between 1000 and 2000 and <i>dpi-pre-detection-max-pkts</i> between 1 and 5. This allows DPI to return its results so service rules are matched. Subsequent packets are dropped once the maximum limit is reached.
46387 Symantec database download may not work on ISM-u.	Symantec database download may not work on ISM-u. Workaround: No workaround exists for this issue.
45392 EDR flow rate is too low.	EDR flow rate is too low (<20,000 per second). Workaround: No workaround exists for this issue.
45322 NAS restarts EdrAgent when a restart is not required.	NAS restarts EdrAgent when a restart is not required. Workaround: For all infrastructure event-log policies that are debug severity and filter on IMSI, use the following command to prevent EdrAgent from logging excessively: <code>(config)# infrastructure logging event-log policy <policy-name> filter-field component filter-value EdrAgent exclude true</code>
40930 In IP non-transparent mode, only one of the s-HD cores handles all downstream Gi packets, which causes a bottleneck.	In IP non-transparent mode, only one of the s-HD cores handles all downstream Gi packets, which causes a bottleneck. Workaround: Provision multiple loopback addresses to ensure that a single s-HD core does not experience a bottleneck. For optimal load distribution, provision as many loopback addresses as there are s-HD cores.

Anomaly	Description
44105 Acuitas may mark a network element as unmanaged.	During OSU, if TACACS is enabled, Acuitas may mark a network element as unmanaged. Workaround: <i>Remanage from Acuitas.</i>

Existing Major Anomalies

The following major anomalies found in an earlier release also exist in this release.

Anomaly	Description
47291 In UDP/QUIC rate limiting, when flow aggregation is enabled, the client bitrate is much lower than the configured bitrate.	In UDP/QUIC rate limiting, when flow aggregation is enabled, the client bitrate is much lower than the configured bitrate. When flow aggregation is disabled, the client bitrate is approximately equal to the configured bitrate. Workaround: <i>Disable QUIC flow aggregation to make the client bitrate approximately equal to the configured bitrate.</i>
45744 Session loss on U-Plane when GwAgent reboots or switches over.	Session loss on the U-Plane when GwAgent reboots or switches over. This occurs for sessions that receive modification requests from the C-Plane before the Fastpath<->Gateway MCD audit is complete. Workaround: <i>No workaround exists for this issue.</i>
46807 Logical bearer ID is used as the bearer ID in Acct-Session-Id.	Logical bearer ID is used as the bearer ID in Acct-Session-Id. Workaround: <i>No workaround exists for this issue.</i>
46636 When PCRF changes default bearer QCI/ARP and installs PCC rules with the same QCI/ARP, a dedicated bearer is created instead of the PCC rules being installed on the default bearer.	When PCRF changes default bearer QCI/ARP through the Default-EPS-Bearer-QoS AVP and installs PCC rules with the same QCI/ARP through either CCA-I (for Gx Rejoin only), CCA-U, or RAR, a dedicated bearer is created instead of the PCC rules being installed on the default bearer. Workaround: <i>Currently, MCC does not support simultaneous changing of default bearer QCI/ARP <u>and</u> the installation of PCC rules on the default bearer. If PCRF sends these separately, MCC does support that call flow.</i>
41498 When validating a username/password via TACACS for a CLI user and an invalid password is detected, non-printable characters may be displayed.	When validating a username/password via TACACS for a CLI user and an invalid password is detected, non-printable characters may be displayed after a valid password is subsequently entered. Workaround: <i>This is a display issue only. After logging in with a valid password, log out and then log in again to clear the issue.</i>
46102 Configuration changes to OSPF parameters are not preserved following a CPM reboot.	Configuration changes to OSPF parameters are not preserved following a CPM reboot. Workaround: <i>When the lead CPM changes, reconfigure the OSPF parameters.</i>

Anomaly	Description
46433 If a CPF loses connectivity to the UPFs for an extended period, it could result in an out of memory condition for the gateway processes.	If a CPF loses connectivity to the UPFs, the CPF gateway processes increases memory consumption by a few bytes for each message not acknowledged by the UPF peer. This could result in an out of memory condition for the gateway processes after an extended period of time if connectivity is not restored and if established sessions continue to be present on the CPF. Workaround: <i>For extended UPF outages, disable periodic statistic fetch from UPF for either EDRs or CDRs.</i>
46428 When a data profile switch occurs and new dynamic rules are installed with the same QCI and ARP, a dedicated bearer is incorrectly created.	When a data profile switch occurs and new dynamic rules are installed with the same QCI and ARP as that of the default bearer, a dedicated bearer is created instead of adding the rule to the existing default bearer. Workaround: <i>No workaround exists for this issue.</i>
46199 PGW supports volume-measurement-count as totalOctets only.	Currently, PGW supports volume-measurement-count as totalOctets only. If configured, inputOctets and outputOctets are also counted as totalOctets. Workaround: <i>No workaround exists for this issue.</i>
46280 tunnel-fragmentation-type in a network-context configuration is applicable to IPv4 only.	The tunnel-fragmentation-type parameter in a network-context configuration is applicable to IPv4 only. This parameter can be configured for IPv4 as inner or outer to specify which IP header should be considered for MCC-initiated fragmentation. This parameter has no effect on IPv6 MCC-initiated fragmentation. Workaround: <i>No workaround exists for this issue.</i>
46299 Interface-level interim timer is not reset on offline charging interface.	When an offline charging interface is configured with both volume threshold and update interval, and multiple bearers have been created, the volume limit per bearer may generate a CDR but it does not reset the interface-level interim timer. Therefore, the next CDR may be generated per the timer which was started at session start. Workaround: <i>No workaround exists for this issue.</i>
45334 When MCC CUPS systems are configured for Lawful Intercept option <i>Call Content from CPF LI Targets</i> , the intercepted packets are not properly delivered to the CPF MCC.	When MCC CUPS systems are configured for Lawful Intercept option <i>Call Content from CPF LI Targets</i>, the intercepted packets are not properly delivered to the CPF MCC. Therefore, X3 events are not reported to LEMF, and the X3 counters on CPF MCC are not incremented. Workaround: <i>Configure the MCC CUPS system for Lawful Intercept option <i>Call Content from UPF Dynamic LI Targets</i> or <i>Call Content from UPF Static LI Targets</i>.</i>
45881 BGP processing may fault due to heartbeat timeout.	MCC has 128 lightweight network-contexts configured which have the same ip-subpool across all network-contexts. In BGP, these are configured as VRF with the same rt-import. Each VRF for a single route in the local VRF table has 128 nexthops, which also applies to all other routes in every VRF. This greatly increases BGP processing and may cause a fault due to heartbeat timeout. Workaround: <i>No workaround exists for this issue.</i>

Anomaly	Description
45690 HTTP proxy task faults/restarts.	HTTP proxy task faults/restarts when both caching and user-space-networking are enabled. Workaround: <i>No workaround exists for this issue.</i>
45775 On ISM, UE's Web page load time is negatively affected.	On ISM, when user-space-networking and proxy are enabled and ip-transparency is disabled, UE's Web page load time is negatively affected. Workaround: <i>No workaround exists for this issue.</i>
45811 FlowAgent/TransProxySipAlg faults when dynamic configuration change occurs from ALG to SIP.	FlowAgent/TransProxySipAlg faults whenever a dynamic configuration change occurs from alg none to alg sip, or from any other alg to alg sip, under the services subscriber-firewall subscriber-firewall-flow firewall_flow tagged to the data profile. This is observed only when there are active flows for UDP/TCP traffic and flows for a particular subscriber and the firewall configuration is changed as described above. Workaround: <i>Do not change ALG configuration under services subscriber-firewall subscriber-firewall-flow firewall_flow from any alg to sip when there are active data flows for that subscriber.</i>
44354 After upgrading to 9.2, file manager purges files in the <i>tosdb/content/contentinsertion</i> and <i>tosdb/content/policylists</i> folders.	In Release 9.2.0.0, the <i>tosdb/content/contentinsertion</i> and <i>tosdb/content/policylists</i> folders were moved to an external storage partition to increase available disk space. After upgrading to 9.2, files formerly in these two folders were purged by file manager according to the maximum age threshold configuration of one year. Workaround: <i>Prior to upgrading to 9.2, in the CLI, ssh to the master MCM on port 2222 and issue the following commands on the master MCM to prevent the files from being purged for one year. A permanent fix will be released in a subsequent 9.2.0.x patch.</i> <pre> root@an3000-mcm-slot7cpu1:~# cd /tosdb/content/policylists/ root@an3000-mcm-slot7cpu1:/tosdb/content/policylists# find -type f -exec ls -l {} + -rw-rw---- 1 content cli-content 0 Jun 5 2017 ./touch.txt -rw-rw---- 1 content cli-content 0 Jun 5 2017 ./url/userlists/touch.txt root@an3000-mcm-slot7cpu1:/tosdb/content/policylists# find -type f -exec touch {} + root@an3000-mcm-slot7cpu1:/tosdb/content/policylists# find -type f -exec ls -l {} + -rw-rw---- 1 content cli-content 0 Apr 26 15:36 ./touch.txt -rw-rw---- 1 content cli-content 0 Apr 26 15:36 ./url/userlists/touch.txt root@an3000-mcm-slot7cpu1:/tosdb/content/policylists# root@an3000-mcm-slot7cpu1:~# cd /tosdb/content/contentinsertion/ root@an3000-mcm-slot7cpu1:/tosdb/content/contentinsertion# find -type f -exec ls -l {} + root@an3000-mcm-slot7cpu1:/tosdb/content/contentinsertion# find -type f -exec touch {} + root@an3000-mcm-slot7cpu1:/tosdb/content/contentinsertion# find -type f -exec ls -l {} + </pre> <i>To recover files after upgrading to 9.2, if the prior release had the same files that are required in the 9.2 release, copy the files from the /boot/partitions/previous/tosdb/content/contentinsertion and /boot/partitions/previous/tosdb/content/policylists folders.</i>
44267 If multiple Virtual Functions (VFs) are assigned on the same Physical Function (PF), the order in which the VFs are sent to the Linux kernel on reboot is inconsistent.	If multiple Virtual Functions (VFs) are assigned on the same Physical Function (PF), the order in which the VFs are sent to the Linux kernel on reboot is inconsistent. This causes MCC to map the interface incorrectly and may lead to communication/routing issues over the CPF, data plane, and for N/S (external) traffic. Workaround: <i>For the ESXi hypervisor, rebooting the VM may clear the issue.</i>

Anomaly	Description
39835 Administratively disabling a NAT pool does not clear the associated session on the UPF.	Administratively disabling a NAT pool does not clear the associated session on the UPF. Workaround: Issue the subscriber clear-local command before disabling the NAT pool to ensure no sessions are left hanging.
43618 Dynamically created interface groups are not deleted for non-CUPS sessions.	Dynamically created interface groups are not deleted for non-CUPS sessions, even when no target is currently using them. Workaround: No workaround exists for this issue.
43795 When a host that has an active MCM, CSM, and SSM is powered off, up to 60 seconds of data loss is observed for sessions hosted on the SSM.	When a host that has an active MCM, CSM, and SSM is powered off, up to 60 seconds of data loss is observed for sessions hosted on the SSM. Workaround: No workaround exists for this issue.
43080 In CUPS environment, SGW/PGW control plane and SAEGW user plane sessions get deleted if push-to-db is enabled.	In CUPS environment, SGW/PGW control plane and SAEGW user plane sessions get deleted if push-to-db is enabled. This is a known limitation; do not enable push-to-db for CUPS. Workaround: No workaround exists for this issue.
42152 Multiple crashes found at the Geo Standby while sync is in progress for the SGW sessions.	Multiple crashes found at the Geo Standby while sync is in progress for the SGW sessions. This is a known limitation; do not enable push-to-db for Geo Redundancy. Workaround: No workaround exists for this issue.
43052 Statistics associated with fragmentation of IPv6 packets are occasionally inconsistent.	Output of the show security statistics data-stats CLI command may occasionally show inconsistent statistics associated with fragmentation of IPv6 packets. Workaround: No workaround exists for this issue.
42132 Commit fails when multiple network-contexts, APNs, and subscriber analyzers are configured within a short amount of time.	When multiple network-contexts, APNs, and subscriber analyzers are configured within a short amount of time, the commit fails with error message Agents not ready . This error is caused by one of the tasks not responding to the validation request sent by the service director. Workaround: Retry the configuration.
41247 Error message when 2K lightweight VRFs are configured with five subpools each, then the subpools are deleted.	Agents not ready error message when 2K lightweight VRFs are configured with five subpools each, then the subpools are deleted in order to recreate the subpools. Workaround: To restart VRF and subpool creation, delete the subpools within the VRF then delete the VRF configuration.
40839 CSM reboot issue for subpools with more than two prefixes.	2K VRFs with 10K subpools is supported when each subpool has two prefixes. Subpools with more than two prefixes may cause an issue with CSM reboots. Workaround: Only subpools with two prefixes is supported.

Anomaly	Description
42413 Rebooting the lead CSM causes OSPF reconvergence to be delayed by approximately one minute.	Rebooting the lead CSM causes OSPF reconvergence to be delayed by approximately one minute. During this time, OSPF learned routes are unreachable and traffic to these destinations is dropped. Workaround: <i>No workaround exists for this issue.</i>
41666 ESXi 6.5 installation issue.	ESXi 6.5 installation does not apply the management network configuration and does not setup networks automatically. Workaround: <i>No workaround exists for this issue.</i>
42180 SSM faults when GRE keepalive is disabled after MCM switchover.	SSM faults when GRE keepalive is disabled after MCM switchover. Workaround: <i>During MCM switchover, do not delete the keepalive configuration. After the switchover is complete, Operators can update the keepalive configuration accordingly.</i>
40920 Zenmate VPN connections are not blocked by MCC.	Zenmate VPN connections are not blocked even though the Operator has applied the block rule on the MCC. This application is not classified properly in the blocking scenarios because it takes different path/pattern to establish the connection, so it is treated as an unknown application. Workaround: <i>No workaround exists for this issue.</i>
42464 FaceTime video call is not blocked by MCC.	FaceTime video call is not blocked even though the Operator has applied the block rule on the MCC. When a blocked FaceTime application reconnects, it uses a different path which is not classified, and continues to function as an unknown application flow. Workaround: <i>No workaround exists for this issue.</i>
40540 Disabling geo-group admin state on an active cluster causes the active cluster to stay in electing mode and call failures.	Disabling geo-group admin state on an active cluster causes the active cluster to stay in electing mode and call failures. Workaround: <i>Perform a failover over to the standby MCC, then admin disable geo-group.</i>
38852 Upgrade fails due to prepare timeout.	Upgrade fails due to prepare timeout. Workaround: <i>Perform a force upgrade: image-management upgrade force <version>.</i>
34851 Control plane processes unresponsive after changing the VLAN tag.	Changing the VLAN configuration on the interfaces may result in the control plane processes (BGP or BFD) becoming unresponsive, resulting in watchdog restarting the process. Functionality after the restart is also non-deterministic. Workaround: <i>If the processes are restarted, the proper workaround is to restart the VM or blade that hosts these processes after the VLAN configuration is modified.</i>
39789 CUPS bearer is not deleted on GTP-U Echo timeout.	CUPS bearer is not deleted on a GTP-U Echo timeout even if clear-bearers-on-gtp-u-path-failure is enabled. Workaround: <i>No workaround exists for this issue.</i>
42545 MCC may fail to forward large fragmented packets.	If very large packets (original size greater than 20K) are sent to MCC in fragments, MCC may fail to forward the packets. Workaround: <i>No workaround exists for this issue.</i>

Anomaly	Description
37203 MCC CPF sends a redundant Sx Deletion Request.	MCC Control Plane Function (CPF) sends a redundant Sx Deletion Request even when a previous Modification Request was responded to with a “No Context Found” message from the User Plane Function (UPF). Workaround: <i>No workaround exists for this issue.</i>
38961 Maximum number of CALEA targets alarm may not be generated.	Under certain circumstances, the anCaleaTargetDBMaxSubsLimitReached alarm is not generated when the maximum number of allowed targets (32,768) are provisioned. Workaround: <i>Issue the command show calea service-status to check if the maximum number of allowed targets has been reached.</i>
40118 If multiple user plane services are configured on a cluster, show subscriber summary displays the wrong user-plane-service.	If multiple user plane services are configured on a cluster, the show subscriber summary command displays the wrong user-plane-service. In this release, only one user-plane-service per MCC cluster is supported. Workaround: <i>No workaround exists for this issue.</i>
37798 BFD all-sessions summary command stops working after OSU when backup MCM takes over after upgrade.	During OSU upgrade, the show ip-protocol bfd all-session-summary command does not display any entries between one MCM upgrade and the time required to complete the entire OSU upgrade. However, the show ip-protocol bfd sessions and show ip-protocol bfd session-detail commands operate without any issues. After the OSU fully completes, the command displays all entries. Workaround: <i>Use the following commands as a workaround:</i> show ip-protocol bfd sessions show ip-protocol bfd session-detail
37780 ASM reboot during OSU.	If OSU is triggered from releases 7.2.4, 7.2.5, 7.2.6, 7.2.7, 7.2.8 then it may cause nodes to reboot. If this occurs on an ASM node while performing an OSU to 8.1 release, then OSU fails. At this point, the only solution is to upgrade the cluster by performing a cluster reboot. Workaround: <i>No workaround exists for this issue.</i>
31773 EW redundancy is not working after bringing down one of EW.	When using vmware host ixgbe driver versions 4.1.1.4-iov and 4.4.1-iov, power cycling a connected 6125 data switch (E/W or N/S ports) may result in host ports not recovering connectivity. This condition affects both SSM and CSM blade types. In addition, rebooting the VMs does not recover and a blade reboot is necessary. Workaround: <i>To avoid this issue, Affirmed Networks recommends using vmware version 3.21.4.3iov of ixgbe host driver.</i>
27624 There is no command to remove users added by TACACS+.	If a remote user ID is added to a cluster using TACACS+, Operators can’t manually remove that user ID from the cluster using a command.

Anomaly	Description
26187 GRE+IPSEC: MTU change on GRE heavyweight interface is not updated at CSM kernel.	Modification of the MTU on GRE heavyweight interface is not updated at CSM kernel. The MTU is set to 1476 in the kernel and there is no option to modify it. This does not affect the behavior of the UE traffic going over the GRE interface; this only limits the MCC originated traffic going over the GRE interface to an MTU of 1476. Workaround: <i>No workaround exists for this issue.</i>
17730 When the shared UE pool feature and the APN mapping feature are both supported for a subscriber, MCC does not force the requirement of the mapped APN to be present in the same network-context as the mapped UE pool.	When the shared UE pool feature and the APN mapping feature are both supported for a subscriber, MCC does not force the requirement of the mapped APN to be present in the same network-context as the mapped UE pool. This may result in the UE IP address being assigned from a different network-context than what is mapped for UE pool sharing and therefore the UE IP address routes are not advertised. Workaround: <i>The network-context of the mapped APNs must point to the correct mapped-network context (for UE pool sharing) from which the UE IP address needs to be allocated.</i>
17789 Shared ue-pool: MCC allows the <code>network-context mapped-ue-pool</code> field to be deleted from a mapped network-context with an existing session.; session persists but the ue-pools for that session to neighboring peer routers is no longer advertised.	MCC allows the <code>network-context mapped-ue-pool</code> field to be deleted from a mapped network-context with an existing session. When the user deletes this field, the session persists, but the ue-pools for that session to neighboring peer routers are no longer advertised. Workaround: <i>Do not remove network-context mapped-ue-pool from the network-context if sessions are currently using that ue pool. If you must perform this task, delete all affected sessions.</i>
9198 Adding route-map to an established BGP neighbor does not result in dynamic update.	If an applied route-map to BGP peer or redistribution is modified, previously advertised routes are not dynamically updated. Workaround: <i>Operators must clear BGP in order to reprocess route updates and capture the new route-map modifications.</i>

Problems Fixed

This section lists the corrections in this release.

Corrections in Release 9.4.0.0

This release includes corrections from the 9.3.2.3, 9.3.1.2, 9.3.0.2, 9.2.2.11, 8.4.2.6, 8.4.1.16, 8.3.3.8, 8.3.2.5, 7.9.4.11, 7.2.16.9, and 7.2.13.6 releases. See the applicable release notes for details.

The following anomalies were corrected in this release:

Anomaly	Description
42344 Attempting to program the multicast MAC address when the port is up and running causes a fault.	Multicast MAC addresses are now programmed when the port is brought up.
47289 In a PFCP Association Release State, if the PFCP Association Release message send fails, the state machine doesn't transition to the other state. This leaves the PFCP Peer state machine hung in the releasing state.	Corrected the logic in the PFCP Association Release State machine transitions.
47169 If any PGW task is killed and restarted after an MCM switch over, it does not accept any more calls until the master PGW task restarts.	Fixed the issue by making the IsAcctOnSent flag persistent, so it restores its value in the standby MCM as well.
47115 The Af-Charging-Id is reported in OFCS data records even if it was not received from the PCRF.	Fixed the issue by resetting the old values in the local service data object.
46923 Whenever the TFTP server initiates an error with error indication (5), these packets fail to reach the UE when ALG is enabled. The client connection might tear down at the client, based on the timeout.	Whenever TFTP server initiates an error with error indication (5), these packets now reach UE. The connection at MCC tears down after a 30-second timeout.
46903 PGW reports a flow as tethered even though the QI value reported in the EDR is less than the configured tethering QI value.	Fixed the issue by adding a subscription to the MCD table for the tethering QI.
46900 A race condition between multiple NAS worker threads may cause NAS task to fault while handling multiple high-throughput sessions.	Fixed the issue by applying mutex locks appropriately.
46899 To add more than 8 VLANs in a Fortville interface, you need to have trust mode enabled.	Affirmed Networks recommends the i40en-18.9.21.1 driver version to set the trust mode persistently. This allows more than 8 VLANs to be added.

Anomaly	Description
46897 Under certain race conditions, NAT connection fails when the round-robin algorithm is selected for subpool selection.	Fixed the issue by keeping the map of sub-pools under NAT-POOL instead of keeping it at the global level.
46855 Generated CDRs do not contain Traffic Data Volume parameters because the fields set to “no” in the interim template are not being written into the CDRs.	Fixed the issue by correcting the check for the entry into the CDRs.
43953 Some MCC files contain sensitive information that is world readable which is a security issue for the system.	Changed all sensitive files to readable/writable only for the file owner and users within the same user group.
47250 Some BFD sessions not restored after reboot when bfd-static-route is configured.	Deprecated the bfd-static-route parameter until a fix is implemented. Instead, use the enable-all-static parameter to enable BFD sessions for all IPv4/IPv6 static routes on the interface. Deprecated the bfd-auth-profile parameter. Support for this functionality will be implemented in a future release.
47207 Account passwords are reset to default after OSU from 9.2.2.6 to 9.2.2.11 and 9.3.2.1 to 9.3.2.3.	The expected host key was not found when the internal ssh user attempted to contact another node as part of the cluster's internal operations. Fixed the issue by preventing the internal user from learning internal host keys.
47318 When CRBN removal from the PCRF causes the last active online rating group to move to the blocked state, the FINAL for the rating group is sent in a CCR-U instead of a CCR-T.	When a CRBN removal from the PCRF causes a rating group to move to the blocked state, a CCR-T is now initiated towards the OCS if this rating group is the last active rating group for a particular subscriber.
47148 Upon reboot, MCM does not release the virtual IP address used to communicate with the payload nodes, causing some nodes to go down after MCM switchover.	Changed the grid's signal handler to only use signal safe functions (for example, <code>execv</code> instead of <code>system</code>).
47109 If the PCRF sends non-zero GBR/MBR values for non-GBR QCI (QCI=5) in ChargingRuleInstall, PGW applies them and also sends a UBR over the GTP interface even though it is a non-GBR bearer. This is default behavior for the PGW. As a result, traffic may be dropped in Fastpath if the MBR value is low, and the GTP endpoint may not expect a UBR with non-zero GBR/MBR values for a non-GBR bearer.	Added a new parameter, ignore-mbr-gbr-for-non-gbr-bearer, to the gateway-common profile. By default, this parameter is disabled. When enabled, the PGW ignores non-zero GBR/MBR values for a non-GBR bearer. A non-GBR bearer is defined by the QCI value (between 5 to 9 or 69 to 70 or 79 to 80). In this case, PGW ignores the non-zero GBR/MBR values received in ChargingRuleInstall, and they will not be applied in Fastpath or GTP. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway profile gateway-common <gateway-profile-name> ignore-mbr-gbr-for-non-gbr-bearer</pre> Ignore non-zero MBR/GBR values for non-Gbr Bearer even if received from PCRF

Anomaly	Description
46550 Occasionally, the extended-storage create and delete commands fail.	Corrected the underlying Linux utility, which is used to partition the disk, to be used interactively.

MCC Security Updates

Nessus security scan reports are available upon request. This release includes the following security updates:

Nessus ID	CVE	Name
123135	CVE-2019-13115	Debian DLA-1730-3 : libssh2 regression update
	CVE-2019-3859	Debian DLA-1730-3 : libssh2 regression update
126512	CVE-2019-13232	Debian DLA-1846-2 : unzip regression update
126607	CVE-2016-9112	Debian DLA-1851-1 : openjpeg2 security update
	CVE-2018-20847	Debian DLA-1851-1 : openjpeg2 security update
126833	CVE-2019-13636	Debian DLA-1856-1 : patch security update
126834	CVE-2019-11719	Debian DLA-1857-1 : nss security update
	CVE-2019-11729	Debian DLA-1857-1 : nss security update
126836	CVE-2018-5743	Debian DLA-1859-1 : bind9 security update
126926	CVE-2016-4609	Debian DLA-1860-1 : libxslt security update
	CVE-2016-4610	Debian DLA-1860-1 : libxslt security update
	CVE-2019-13117	Debian DLA-1860-1 : libxslt security update
	CVE-2019-13118	Debian DLA-1860-1 : libxslt security update
126964	CVE-2019-10639	Debian DLA-1862-1 : linux security update
	CVE-2019-13272	Debian DLA-1862-1 : linux security update
	CVE-2019-2101	Debian DLA-1862-1 : linux security update
127076	CVE-2019-13638	Debian DLA-1864-1 : patch security update

Documentation Updates and Corrections

This section lists any new features, documentation updates, and corrections that were not included in the released documentation set.

Mobile Content Cloud Maximum Supported Object Instances

The attached Adobe Acrobat document lists the maximum number of supported Mobile Content Cloud object instances.

[Click this link to view the attached document.](#)

Correction to IP-based Fingerprinting for Detection of NAT

The *Affirmed Networks System Administration Guide* incorrectly describes *IP-based Fingerprinting for Detection of NAT* as supported on Windows systems up to Windows 7. This method of detection is not supported in this release.

Configuring the Node ID in CDRs

[Table 6](#) summarizes the MCC behavior when using the `node-id` parameter under the gateway service and the `unique-cdr-nodeid` parameter under the offline-charging interface together.

Table 6 Configuring the `node-id` and `unique-cdr-nodeid` parameters together

zone default gateway [sgw pgw ggsn] <gateway-service-name> node-id <node-id-string>	service-construct interface offline-charging <offline-charging-name> [pgw sgw ggsn] unique-cdr-nodeid [enabled disabled]	Value of Node ID in CDR
<node-id-string> configured	enabled	<node-id-string>_Service-Type-ID
<node-id-string> not configured	enabled	<gateway-service-name>_Service-Type -ID
<node-id-string> not configured	disabled	<gateway-service-name>
<node-id-string> configured	disabled	<node-id-string>

Lawful Intercept: Multiple X2/X3 Delivery Destinations

In this release, MCC supports the Lawful Intercept Multiple X2/X3 Delivery Destinations feature for both persistent and non-persistent targets.

In previous releases, the show calea target-status command displayed the following fields:

Only for persistent targets:

- x2-interface-group-name
- x3-interface-group-name

Only for non-persistent targets:

- x2-interface-name
- x3-interface-name

In this release, the show calea target-status command now displays all of the fields according to the follow rules:

When non-persistent targets are associated directly with an interface and no Multiple HI2/HI3 configuration is present:

- x2-interface-name - populated
- x3-interface-name - populated (for iri-plus-cc targets), otherwise displays empty string ""
- x2-interface-group-name - displays empty string ""
- x3-interface-group-name - displays empty string ""

When non-persistent targets are used with the Multiple HI2/HI3 feature configured:

- x2-interface-name - displays empty string ""
- x3-interface-name - displays empty string ""
- x2-interface-group-name - populated
- x3-interface-group-name - populated (for iri-plus-cc targets with Multiple HI3 LEMF once the dynamic X3 group has been created), otherwise displays empty string ""

Update to Configuration Options for cluster <id> node <id> admin-state

The **disabled** and **maintenance** options for **cluster <id> node <id> admin-state** described in the *Affirmed Networks System Administration Guide*, are not supported and should not be used.

The following use case examples describe how to perform maintenance actions on the cluster, and increase and reduce system capacity.

Performing Maintenance Actions

- 1 Use the hypervisor's VM shutdown procedure to sequentially power off each VM running on the host.
- 2 Wait for redundancy to transfer service to the standby VMs and for the system to stabilize.
- 3 Perform the required maintenance on the host.
- 4 Sequentially power up the VMs, making sure that the system stabilizes after each VM powers up.

Increasing/Decreasing System Capacity

CAUTION: The following procedure impacts service; either move service off the MCC or perform these actions during periods of decreased traffic.

- 1 Sequentially enable or disable the VMs.
- 2 Reboot the cluster.
- 3 Transfer service back to the MCC.